
Analysis

UCLA Math-131A Lecture notes

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1 Lecture 1

The integers and induction

Definition. A statement (or proposition, or assertion, or claim) is a declarative sentence that may be true or false.

Examples. We sometimes denote the statements by capital letters P, A, X, Y or $P_1, P_2, \dots, P_n$.

- $3 \leq 4$
- Any two lines in a plane intersect exactly once
- All dogs are grey

Some symbols.

- The symbol $\{\}$ means a “set” and the object in the set is called an “element”.
- The symbol $\in$ means “is an element of” or “belongs to” or “is contained in”.
- The symbol $\forall$ means “for all” or “for any”.
- The symbol $\exists$ means “there exists”.
- The symbol $\subseteq$ means “is a subset of” and the symbol $\supseteq$ means “is a superset of”. For example, $A \subseteq B$ means all elements in A are also in B .
- The symbol $X \implies Y$ means “the statement X implies the statement Y ”. The symbol $X \impliedby Y$ means “the statement X is implied by the statement Y ”.
- The symbol $X \iff Y$ means “the statement X holds if and only if the statement Y holds” or “the statement X is equivalent to the statement Y ”.

Definition. We define the following sets

- $\mathbb{N} = \{1, 2, 3, \dots\}$ (the natural numbers)
- $\mathbb{Z} = \{0, 1, -1, 2, -2, \dots\}$ (the integers)
- $\mathbb{Q} = \{\frac{k}{m} : k, m \in \mathbb{Z}, m \neq 0\}$ (the rational numbers)

Furthermore, we temporarily define $\mathbb{R}$ the set of all “real numbers”. The notation $\mathbb{Q}$ is understood as “the collection of objects of the form $\frac{k}{m}$, where $k, m \in \mathbb{Z}$ and $m \neq 0$ ”. It is readily checked that $\mathbb{N} \subseteq \mathbb{Z} \subseteq \mathbb{Q}$, i.e., every natural number is an integer and every integer is a rational number.

Warning. The definition of $\mathbb{N}, \mathbb{Z}$, and $\mathbb{Q}$ are rigorous and mathematical. However, the definition of real numbers $\mathbb{R}$ is not rigorous, since we don’t have a formal (mathematical) way of describing real numbers yet. Our first big goal is to provide a rigorous description of “real numbers”.

First we want to answer to the following question:

What is the set of real numbers $\mathbb{R}$ and how can we describe it rigorously?

We will build towards an answer to the question. The most important defining property of $\mathbb{R}$ is the **Completeness Axiom**. This is also what distinguished the set of rationals $\mathbb{Q}$ from the set of real numbers $\mathbb{R}$: $\mathbb{Q}$ doesn't have the Completeness Axiom while $\mathbb{R}$ has the Completeness Axiom.

We assume some familiarity of the set of natural numbers $\mathbb{N} = \{1, 2, 3, \dots\}$. For example, how the operations $+$, $\cdot$, and the ordering $\leq$ works with $\mathbb{N}$. Besides, we state the following property about $\mathbb{N}$.

Well-Ordering Principle. Suppose $S \subseteq \mathbb{N}$ is non-empty. Then S has a least element, i.e., there is some $a \in S$ such that for all $b \in S$, $a \leq b$.

The WOP seems quite obvious but actually we cannot prove it with just the familiar properties that the natural numbers satisfy under addition and multiplication. Hence, we shall regard WOP as an axiom.

Interestingly though, the well-ordering principle of $\mathbb{N}$ is equivalent to the Principle of Mathematical Induction.

Principle of Induction. Let $n \in \mathbb{N}$ and suppose that P_n is a statement. Suppose that

- (1) P_1 holds (this is called the “base case for the induction”), and
- (2) for every $n \in \mathbb{N}$, if P_n holds, then P_{n+1} holds (this is called the “inductive step”).

Then P_n holds for every natural number $n \in \mathbb{N}$.

Proof. Let $S := \{n \in \mathbb{N} : P_n \text{ is false}\} \subseteq \mathbb{N}$. For the sake of contradiction, we assume that P_n does not hold for every natural number $n \in \mathbb{N}$. Thus, $S \neq \emptyset$. By the well-ordering principle, the set S has a least element $m \in \mathbb{N}$. Since P_1 holds by assumption, we know that $1 < m$ and thus $m - 1 \in \mathbb{N}$. Since m is the least element of S , we infer that the natural number $m - 1 \notin S$ and P_{m-1} is true. The assumption (2) implies P_m is true. Contradiction! $\square$

Notation. Suppose $M, N \in \mathbb{Z}$ are integers such that $M \leq N$ and suppose that we are given numbers $a_M, a_{M+1}, \dots, a_N$ indexed by all integers between M and N . We denote the finite summation of the integers $a_M, \dots, a_N$ by

$$\sum_{k=M}^N a_k := a_M + a_{M+1} + \dots + a_N.$$

The “ k ” in the expression $\sum_{k=M}^N a_k$ is referred to as the index of summation and it is a dummy variable, i.e., a variable that has no actual significance and can be replaced by other index. This is analogous to the control variable of a *for-loop* in a computer program.

2 Lecture 2

Example. Prove that the equality

$$\sum_{k=1}^n k = \frac{n(n+1)}{2}$$

holds for all $n \in \mathbb{N}$.

Proof. We aim to prove this by induction on $n \in \mathbb{N}$. Let P_n be the statement that

$$\sum_{k=1}^n k = \frac{n(n+1)}{2}.$$

We first show that P_1 holds. The assertion P_1 says that

$$\sum_{k=1}^1 k = \frac{1(1+1)}{2}.$$

The left-hand side is clearly 1, and the right-hand side is also 1. Thus, the statement P_1 holds.

Let $n \in \mathbb{N}$. We assume that P_n holds, i.e.,

$$\sum_{k=1}^n k = \frac{n(n+1)}{2}.$$

We must show that P_{n+1} also holds. That is, we need to show:

$$\sum_{k=1}^{n+1} k = \frac{(n+1)(n+2)}{2}.$$

We can write the left-hand side as:

$$\sum_{k=1}^{n+1} k = \sum_{k=1}^n k + (n+1) = \frac{n(n+1)}{2} + (n+1),$$

by induction hypothesis. Thus, we have

$$\sum_{k=1}^{n+1} k = \frac{(n+1)(n+2)}{2},$$

i.e., P_{n+1} holds. By the principle of induction, we conclude that the formula holds for all $n \in \mathbb{N}$. $\square$

We have the following variant of the Principle of Induction, which starts at some integer other than 1.

Principle of Induction starting at N . Let $N \in \mathbb{Z}$ and $\{P_N, P_{N+1}, P_{N+2}, \dots\}$ be statements. Suppose that

- (1) P_N holds, and
- (2) for every integer $n \geq N$, if P_n holds, then P_{n+1} holds.

Then P_n holds for every integer $n \geq N$.

Proof. We will prove this by reducing it to the original principle of induction by shifting. Let $n \in \mathbb{N}$. Let Q_n be the statement that “ P_{n+N-1} holds”. Then, the condition (1) implies that Q_1 holds, since P_N is assumed to hold. The condition (2) implies that for every $n \geq 1$, if Q_n holds, then Q_{n+1} holds, because if P_{n+N-1} holds, then P_{n+N} must also hold.

Thus, Q_n is true for all $n \in \mathbb{N}$ by the Principle of Induction. In other words, P_{n+N-1} holds for all $n \in \mathbb{N}$, which implies that P_n holds for all $n \geq N$. $\square$

The rational numbers

Recall that the set of integers $\mathbb{Z} = \{0, 1, -1, 2, -2, \dots\}$ and the set of rational numbers $\mathbb{Q} = \{\frac{k}{m} : k, m \in \mathbb{Z}, m \neq 0\}$. In a certain vague sense, the rational numbers $\mathbb{Q}$ are incomplete, i.e., there are many theoretical numbers that shall exist, but don't exist as rational numbers.

Remark. Consider a square with sides having length one. Let d be the length of the diagonal. By Pythagorean theorem, we know $1^2 + 1^2 = d^2$. Thus, there is a positive length whose square is 2, which we denote as $\sqrt{2}$. We aim to prove that $\sqrt{2}$ is not a rational number.

Proposition. Any non-zero rational number can be written in a reduce form $\frac{p}{q}$, where p and q are non-zero and the greatest common divisor of p and q is 1.

The proof of the proposition uses the prime factorization from elementary number theory. We omit it here.

Corollary. Any non-zero rational number can be written as $\frac{p}{q}$, where p is odd or q is odd.

Proof. Let $r \in \mathbb{Q}$ be a non-zero rational number and $\frac{p}{q}$ be its reduced form. For the sake of contradiction, we assume that p and q are even integers. Then a common divisor of p and q is 2, contradicting with the reduced form. $\square$

Proposition. There is no rational number $r \in \mathbb{Q}$ with the property that $r^2 = 2$. Consequently, $\sqrt{2}$ is not a rational number.

Proof. For the sake of contradiction, we assume that there is a number $r \in \mathbb{Q}$ such that $r^2 = 2$. It is clear that r is non-zero. By the corollary, we write

$$r = \frac{p}{q},$$

where p is odd or q is odd. We take the square of both sides to obtain

$$r^2 = \frac{p^2}{q^2} \implies 2q^2 = p^2.$$

Since the left-hand side is even, the right-hand side must also be even. Thus, p is even and we may write $p = 2k$, where $k \in \mathbb{Z}$. We plug it back in, yielding

$$2q^2 = (2k)^2 = 4k^2 \implies q^2 = 2k^2.$$

Since the right-hand side is even, we infer that the left-hand side is also even. Thus, q is even. Both p and q are even, contradiction! $\square$

Motivation. We have just proved that $\sqrt{2}$ is not a rational number and thus the set of rationals has “gaps”. This is one big reason that $\mathbb{Q}$ is not adequate for mathematical purpose and we aim to build a better number system that includes numbers such as $\sqrt{2}$.

3 Lecture 3

Inequalities

Definition. For $a \in \mathbb{R}$, we define the absolute value

$$|a| = \begin{cases} a, & \text{if } a \geq 0, \\ -a, & \text{if } a < 0. \end{cases}$$

Proposition. Suppose $a, b \in \mathbb{R}$. Then the following properties hold

(1) $|a| \geq 0$

(2) $|ab| = |a| \cdot |b|$

Proof. (1) If $a \geq 0$, then $|a| = a \geq 0$. If $a < 0$, then $|a| = -a > 0$.

(2) *Case 1:* Suppose $a \geq 0$ and $b \geq 0$. Then $a \cdot b \geq 0$, so $|a \cdot b| = a \cdot b = |a| \cdot |b|$.

Case 2: Suppose $a < 0$ and $b < 0$. Then $-a, -b > 0$, so $a \cdot b = (-a) \cdot (-b) > 0$, hence $|a \cdot b| = a \cdot b = |a| \cdot |b|$.

Case 3: Suppose $a \geq 0$ and $b < 0$. Then $-b \geq 0$, and $a \cdot (-b) \geq 0$, hence $|a| \cdot |b| = a \cdot (-b) = -(a \cdot b) = |a \cdot b|$.

Case 4: Suppose $a < 0$ and $b \geq 0$. This case is similar to Case 3. □

Proposition. Let $a, b \in \mathbb{R}$ and $b \geq 0$. Then

$$|a| \leq b \iff -b \leq a \leq b$$

The proof uses case analysis which is omitted here. The following inequality is perhaps the most important and fundamental inequality in analysis. It will be used frequently throughout this class, sometimes without explicit mention.

Triangle Inequality. For any $a, b \in \mathbb{R}$, we have

$$|a + b| \leq |a| + |b|.$$

Proof. Since $|a| \leq |a|$, we deduce from the proposition that

$$-|a| \leq a \leq |a|.$$

Similarly, $-|b| \leq b \leq |b|$. By adding the two inequalities, we obtain $-|a| - |b| \leq a + b \leq |a| + |b|$. From the proposition again it follows that $|a + b| \leq |a| + |b|$. □

Corollary. For any $a, b, c \in \mathbb{R}$, we have

$$|a - c| \leq |a - b| + |b - c|.$$

Proof. We apply the triangle inequality to $a - b$ and $b - c$,

$$|a - c| = |a - b + b - c| \leq |a - b| + |b - c|.$$

The proof is thus completed. $\square$

Reverse Triangle Inequality. For any $a, b \in \mathbb{R}$, we have

$$|a - b| \geq ||a| - |b||.$$

Proof. This is left as homework exercise. $\square$

The real number system and the Completeness Axiom

In this section, we will discuss the Completeness Axiom for $\mathbb{R}$ which is assumed throughout the rest of the course. This is the axiom that assures us that $\mathbb{R}$ has no “gaps”. Recall that we temporarily denote the set of real numbers by $\mathbb{R}$.

Definition. Let $S \subseteq \mathbb{R}$ be nonempty.

- (1) If a real number $s_0 \in S$ satisfies $s \leq s_0, \forall s \in S$, then s_0 is called the maximum of S and we write $s_0 = \max S$.
- (2) If a real number $s_0 \in S$ satisfies $s \geq s_0, \forall s \in S$, then s_0 is called the minimum of S and we write $s_0 = \min S$.

Examples.

- Consider $S = \{1, 2, 3\}$. Then $\min S = 1$ and $\max S = 3$.
- Let $n \in \mathbb{N}$ and $S = \{s_1, s_2, \dots, s_n\} \subseteq \mathbb{R}$. Then S has a maximum and minimum. Proof follows from induction on the number of elements in S .
- For the closed interval $S = [0, 1]$, $\min S = 0$ and $\max S = 1$. However, the open interval $(0, 1)$ has neither minimum nor maximum. Note that both closed and open intervals are sets that have infinitely many elements.
- The set of positive integers $\mathbb{N}$ has $\min \mathbb{N} = 1$, but no maximum.

Example. The set $\mathbb{Z}$ has neither a minimum nor a maximum.

Proof. For the sake of contradiction, we assume that $\mathbb{Z}$ has a maximum. Denote $a = \max \mathbb{Z}$. Since $a \in \mathbb{Z}$, we certainly also have $b = a + 1 \in \mathbb{Z}$. Since $b > a$ and $b \in \mathbb{Z}$, we derive a contradiction to the maximality of a .

A similar proof by contradiction shows that $\mathbb{Z}$ has no minimum. $\square$

4 Lecture 4

Definition. Let $S \subseteq \mathbb{R}$ be nonempty.

- (1) If a real number M satisfies $M \geq s, \forall s \in S$, then M is called an upper bound of S and we say that S is bounded from above.
- (2) If a real number m satisfies $m \leq s, \forall s \in S$, then m is called a lower bound of S and we say that S is bounded from below.
- (3) We say that S is bounded if it is bounded from above and from below.

Example. Consider $S = (0, 1)$. Then 0 is a lower bound and 1 is an upper bound.

Definition. Let $S \subseteq \mathbb{R}$ be nonempty.

- (1) If S is bounded from above and has a least upper bound M , then M is called the supremum of S and we write $M = \sup S$.
- (2) If S is bounded from below and has a greatest lower bound m , then m is called the infimum of S and we write $m = \inf S$.

Example. Consider $S = (0, 1)$. Prove that 1 is the least upper bound of S and 0 is the greatest lower bound of S .

Proof. Let M be an upper bound of S . By definition, $M \geq s, \forall s \in (0, 1)$. In particular, $M \geq 0$. We aim to prove that $M \geq 1$. For the sake of contradiction, we assume that $M < 1$. There exists a number $M' \in (M, 1) \subseteq (0, 1)$. This contradicts with the fact that M is an upper bound of $S = (0, 1)$. Hence, we have $M \geq 1$. Since 1 is an upper bound of S and $M \geq 1$ for any upper bound M , we deduce that 1 is the least upper bound of S .

A similar proof shows that 0 is the greatest lower bound of S . □

From the previous example, we see that, unlike $\max S$ and $\min S$, the supremum and infimum need not belong to S .

Example. The maximum of a set S (if it exists) is the supremum; the minimum of a set S (if it exists) is the infimum.

Proof. Let $s_0 = \max S$. By definition, $s_0 \geq s, \forall s \in S$. Hence, s_0 is an upper bound of S . Suppose that s_1 is another upper bound of S . By definition, $s_1 \geq s_0$ since $s_0 \in S$. By the arbitrariness of s_1 , we deduce that s_0 is the least upper bound and thus the supremum.

A similar proof shows that the minimum of a set S (if it exists) is the infimum. □

Owing to the following axiom, the supremum of a set exists under very general conditions, as opposed to the maximum.

Completeness Axiom. Every nonempty subset of $\mathbb{R}$ which is bounded from above has a least upper bound. In other words, given nonempty $S \subseteq \mathbb{R}$ and S has at least one upper bound, then $\sup S$ exists.

The defining property of an ordered field is A1–A4 (algebraic), M1–M4 (algebraic), DL (distributive law), and O1–O5 (ordering structure). An example of an ordered field is $\mathbb{Q}$ with the usual operations and order.

Answer to the question “what is the set of real numbers $\mathbb{R}$ ”. In our class, we simply define $\mathbb{R}$ as an ordered field which has the Completeness Axiom. Moreover, $\mathbb{R}$ contains $\mathbb{Q}$ as a subfield. The members of $\mathbb{R}$ are called real numbers.

For reader’s interest, there are several equivalent ways of defining the real numbers $\mathbb{R}$.

1. Axiomatic definition from the unique Dedekind-complete ordered field.
2. Dedekind cuts.
3. Equivalence classes of Cauchy sequences in $\mathbb{Q}$.

To develop $\mathbb{R}$ properly, it would take us several (tedious) lectures. This would defeat the purpose of the course, which is to practice writing rigorous proofs and study some important properties of $\mathbb{R}$ and functions defined on $\mathbb{R}$. A good reference for constructing the real numbers is the book “Principles of Mathematical Analysis” by W. Rudin. The book also goes by the nickname “baby rudin” to distinguish it from Rudin’s more advanced books.

5 Lecture 5

Notation. We denote $\mathbb{I} = \mathbb{R} \setminus \mathbb{Q}$, the set of irrationals.

There is also an equivalent “infimum version” of the Completeness Axiom.

Completeness Axiom (infimum version). Every nonempty subset S of $\mathbb{R}$ which is bounded from below has a greatest lower bound.

Definition. Given a set $S \subseteq \mathbb{R}$, we define $-S$ to be the reflection of S about the origin, i.e.,

$$-S := \{-s : s \in S\}.$$

For example, $-[0, 1) = (-1, 0]$. Note that $-(-S) = S$.

Proof of the Completeness Axiom (infimum version). Since S is bounded from below, there exists some $m \in \mathbb{R}$ such that $m \leq s, \forall s \in S$. This implies that $-m \geq -s, \forall s \in S$. By the definition of $-S$, we deduce that $-m \geq t, \forall t \in -S$. Hence the set $-S$ is bounded from above by $-m$. It is clear that $-S$ is nonempty. By Completeness Axiom, $\sup(-S)$ exists.

Denote $s_0 = \sup(-S)$. We aim to prove that $\inf(S) = -\sup(-S) = -s_0$, i.e.,

$$-s_0 \leq s, \forall s \in S.$$

and

$$\text{if } m' \text{ is a lower upper of } S, \text{ then } m' \leq -s_0.$$

The first claim is equivalent to “ $-s \leq s_0, \forall s \in S$ ”. This follows from the fact that s_0 is an upper bound of $-S$.

For the second claim, let m' be a lower bound of S . We notice that $m' \leq s, \forall s \in S$ implies that $-s \leq -m', \forall s \in S$. It follows that $-m'$ is an upper bound of $-S$. Since s_0 is the least upper bound of $-S$, we infer that $-m' \geq s_0$ which implies that $m' \leq -s_0$.

Therefore, we have proved that $-s_0$ is the greatest lower bound of S , i.e., $-s_0 = \inf S$. $\square$

The following property really requires us to use the Completeness Axiom.

Archimedean property. If $a > 0$ and $b > 0$, then for some positive integer n , we have $na > b$.

One may wonder, do we really need to use such a powerful axiom (Completeness Axiom) to prove this basic property? The answer is yes. In fact, there exists an ordered field that fails the Archimedean property, which is also known as the non-Archimedean ordered field. We refer the proof of Archimedean property to the textbook.

Proposition. (The set $\mathbb{Q}$ is dense in $\mathbb{R}$) Let $a, b \in \mathbb{R}$ with $a < b$. Then there exists a rational $r \in \mathbb{Q}$ such that $r \in (a, b)$.

Proof. Since $b - a > 0$, the Archimedean property shows that there exists some $n \in \mathbb{N}$ such that

$$n(b - a) > 1 \implies nb - na > 1.$$

From here, it is evident that there is an integer m between na and nb . As a result, we find a rational number $r = \frac{m}{n} \in \mathbb{Q}$ such that $a < r < b$.

To find such m explicitly, we argue as follows. By Archimedean property, there exists some positive integer $k > \max\{|an|, |bn|\}$ so that

$$-k < an < bn < k.$$

Let $E = \{j \in \mathbb{Z} : -k \leq j \leq k\}$ and $F = \{j \in E : an < j\}$. It is clear that E and F are finite sets and they are nonempty, since they both contain k . Let $m = \min F$. Then $-k < an < m$. Since the integer $m \in E$ and $m > -k$, we infer that $m - 1 \in E$. It follows that $an < m - 1$ is false by the minimality of m of F . Thus, $an \geq m - 1$ and $m - 1 \leq an < bn - 1$. Since $an < m < bn$, we find explicitly an integer $m \in (an, bn)$. $\square$

Logic

We have introduced and encountered some symbols such as $\exists$ and $\forall$. We will discuss more complicated logical relations.

Definition. The logical negation of a statement A , denoted by $\neg A$, is the statement “not A ”. The statements A and $\neg A$ have exactly one true statement and one false statement.

Remark. The method of proof by contradiction is built on the logical negation. Suppose we want to prove that the statement X holds. We assume that $\neg X$ holds and are able to derive a contradiction. Hence, $\neg X$ is a false statement. Since X and $\neg X$ have exactly one true and one false statement, we deduce that X is a true statement.

Remark. For any statement A , we have $\neg\neg A = A$.

Example. A is the statement “all dogs are grey”, then $\neg A$ is the statement “some dogs are not grey”.

Let X and Y be statements in the following table.

Statement	Negation
X or Y	$\neg X$ and $\neg Y$
X and Y	$\neg X$ or $\neg Y$
$X \implies Y$	X and $\neg Y$
For all x , $A(x)$	There exists x such that $\neg A(x)$
There exists x such that $A(x)$	For all x , $\neg A(x)$

Example. The Archimedean property can be formulated as

$$“\forall a > 0, \forall b > 0, \exists n \in \mathbb{N} \text{ such that } na > b”.$$

The negation of it is

$$“\exists a > 0, \exists b > 0, \text{ such that } \forall n \in \mathbb{N} \text{ we have } na \leq b”.$$

This is what the proof of Archimedean property assumes at the beginning, with the aim of deriving a contradiction.

6 Lecture 6

Proposition. Let $S \subseteq \mathbb{R}$ be nonempty and bounded from above. Let $M = \sup S$. Then for any $M' < M$, there exists some $a \in S$ such that

$$M' < a \leq M.$$

Proof. We aim to prove the first inequality, i.e., $\forall M' < M, \exists a \in S$ such that $M' < a$. For the sake of contradiction, we assume that $\exists M' < M$, such that $\forall a \in S$, we have $M' \geq a$. Then M' is an upper bound of S . Since M is the least upper bound, we infer $M' \geq M$. This contradicts with $M' < M$.

After the first inequality is established, the second inequality follows from the facts that $a \in S$ and M is an upper bound of S . $\square$

The symbols $+\infty$ and $-\infty$

In this section, we discuss two new symbols $-\infty$ and $+\infty$ (usually abbreviated to ∞). They are not real numbers but they are useful symbols for notational purposes. For example, it is convenient to denote the unbounded intervals using $\pm\infty$

$$[a, +\infty) := \{x \in \mathbb{R} : a \leq x\}, \quad (a, +\infty) := \{x \in \mathbb{R} : a < x\},$$

$$(-\infty, b] := \{x \in \mathbb{R} : x \leq b\}, \quad (-\infty, b) := \{x \in \mathbb{R} : x < b\}.$$

Recall that the Completeness Axiom states that if $S \subseteq \mathbb{R}$ is nonempty and bounded from above, then $\sup S$ exists as a real number. We wish to extend the meaning of supremum to sets that are not bounded from above.

Definition. For nonempty $S \subseteq \mathbb{R}$, if S is not bounded from above, then we declare

$$\sup S := +\infty,$$

Likewise, if S is not bounded from below, then we declare

$$\inf S := -\infty.$$

Remark. In the above definition, ∞ is not a real number. Rather, we should interpret the definition as “we all agree that if S is not bounded from above, then we say $\sup S$ is arbitrarily large or larger than any real number”. To make it short, we write $\sup S = \infty$. Now, for any nonempty $S \subseteq \mathbb{R}$, we have

$$\sup S = \begin{cases} \text{the least upper bound of } S, & \text{if } S \text{ is bounded from above,} \\ \infty, & \text{otherwise.} \end{cases}$$

Similar consideration also applies to $\inf S$. There is nothing deep or special about $\pm\infty$. We primarily introduce it because it makes certain commonly occurring statements and expressions shorter.

Limits of sequences

A sequence of real numbers is a function whose domain is $\mathbb{N}$ and codomain is $\mathbb{R}$. The sequence is denoted by $\{a_n\}_{n \in \mathbb{N}}$ or $(a_n)_{n \in \mathbb{N}}$. For each $n \in \mathbb{N}$, $a_n \in \mathbb{R}$ is the output. Occasionally, the domain is $\{n \in \mathbb{Z} : n \geq m\}$ for some $m \in \mathbb{Z}$ and we denote such sequence by $(a_n)_{n \geq m}$. If the domain of a sequence is clear from the context, then we simply write (a_n) .

Examples.

- $a_n = \frac{1}{n^2}, n \in \mathbb{N}$ is the sequence: $\{1, \frac{1}{4}, \frac{1}{9}, \frac{1}{16}, \dots\}$
- $((-1)^n)_{n \geq 0}$ is the sequence: $\{1, -1, 1, -1, \dots\}$

We may have learned the idea of a sequence (a_n) converging to a number $a \in \mathbb{R}$ in Calculus II. We require that as n gets larger and larger, the distance between a_n and a gets smaller and smaller. Now we will quantitatively characterize this concept.

Definition. Let $(a_n)_{n \in \mathbb{N}}$ be a sequence of real numbers. We say that $(a_n)_{n \in \mathbb{N}}$ converges to a real number a if

$$\forall \epsilon > 0, \exists N \in \mathbb{N} \text{ such that } n \geq N \implies |a_n - a| < \epsilon. \quad (*)$$

The number $a \in \mathbb{R}$ is called the limit of the sequence and we write $(a_n)_{n \in \mathbb{N}} \rightarrow a$.

7 Lecture 7

A discussion about proofs

Informal discussion. For any $\epsilon > 0$, there exists an integer N (usually depends on ϵ inversely proportionally) such that the tail starting from N is ϵ -close to a . Although we have written the definition for any $\epsilon > 0$, it is traditionally intuited that ϵ takes the role of a small positive number.

Example. Prove that $(\frac{1}{\sqrt{n}})_{n \in \mathbb{N}} \rightarrow 0$.

Proof. Let $\epsilon > 0$. We need to find $N \in \mathbb{N}$ such that

$$\left| \frac{1}{\sqrt{n}} - 0 \right| < \epsilon, \quad \forall n \geq N.$$

Let N be an integer such that $N > \frac{1}{\epsilon^2}$. Then for any $n \geq N$, we have

$$n \geq N > \frac{1}{\epsilon^2} \implies n > \frac{1}{\epsilon^2} \implies \frac{1}{n} < \epsilon^2 \implies \left| \frac{1}{\sqrt{n}} \right| < \epsilon.$$

By definition, we have proven that $(\frac{1}{\sqrt{n}})_{n \in \mathbb{N}} \rightarrow 0$. □

Definition. We say that a sequence (a_n) converges if $\exists a \in \mathbb{R}$ such that (*) holds. A sequence that does not converge is said to diverge.

Remark. We will prove in a homework exercise that if a sequence converges, then the limit is unique.

Example. Prove that $((-1)^n)_{n \in \mathbb{N}}$ diverges.

Proof. For the sake of contradiction, we assume that the sequence converges. Then there exists some $a \in \mathbb{R}$ such that $((-1)^n)_{n \in \mathbb{N}} \rightarrow a$. Let $\epsilon = 1$. There exists some $N \in \mathbb{N}$ such that

$$|(-1)^n - a| < 1, \quad \text{for all } n \geq N.$$

This leads to $|1 - a| < 1$ and $|1 + a| < 1$. By the triangle inequality, we have

$$|1 - a| + |1 + a| \geq |1 - a + 1 + a| = 2.$$

On the other hand, $|1 - a| + |1 + a| < 1 + 1 = 2$. This is an absurdity. □

Example. Prove that $(\frac{2n+4}{5n+2})_{n \in \mathbb{N}} \rightarrow \frac{2}{5}$.

Proof. Let $\epsilon > 0$. We need to find $N \in \mathbb{N}$ such that

$$\left| \frac{2n+4}{5n+2} - \frac{2}{5} \right| < \epsilon, \quad \forall n \geq N.$$

We compute

$$\frac{2n+4}{5n+2} - \frac{2}{5} = \frac{5(2n+4) - 2(5n+2)}{5(5n+2)} = \frac{16}{5(5n+2)}.$$

Hence, we need to find $N \in \mathbb{N}$ such that

$$\left| \frac{16}{5(5n+2)} \right| < \epsilon, \quad \forall n \geq N.$$

Let N be an integer such that $N > \frac{16}{25\epsilon}$. Then for any $n \geq N$, we have

$$\frac{16}{5(5n+2)} \leq \frac{16}{5(5N+2)} < \frac{16}{25N} < \frac{16}{\frac{16}{\epsilon}} = \epsilon,$$

as desired. □

8 Lecture 8

A discussion about proofs

We provide in the following example a classical technique in analysis, which is to treat the head and tail of a sequence separately.

Example. Let $(s_n)_{n \in \mathbb{N}}$ be a convergent sequence such that $s_n > 0$ for all $n \in \mathbb{N}$. Suppose that $(s_n)_{n \in \mathbb{N}} \rightarrow s$, where $s > 0$. Prove that $\inf\{s_n : n \in \mathbb{N}\}$ exists as a positive real number.

Proof. The set $\{s_n : n \in \mathbb{N}\}$ has a lower bound 0 since $s_n \geq 0, \forall n \in \mathbb{N}$. By completeness axiom, it admits a greatest lower bound. Let $\epsilon = \frac{s}{2} > 0$. There exists some $N \in \mathbb{N}$ such that

$$|s_n - s| < \frac{s}{2}, \quad \forall n \geq N.$$

It follows that $s/2 < s_n < 3s/2$, for any $n \geq N$. Let $m = \min\{s_1, s_2, \dots, s_{N-1}, \frac{s}{2}\}$. It is clear that $m > 0$ since each element is positive. Also, m is a lower bound of $\{s_n : n \in \mathbb{N}\}$ since

$$m \leq s_n, \quad \forall n = 1, \dots, N-1$$

and

$$m \leq \frac{s}{2} < s_n, \quad \forall n \geq N.$$

Hence, the greatest lower bound $\inf\{s_n : n \in \mathbb{N}\} \geq m > 0$. □

Limit theorems for sequences

Definition. We say that the sequence $(s_n)_{n \in \mathbb{N}}$ is bounded if the set $\{s_n : n \in \mathbb{N}\} \subseteq \mathbb{R}$ is bounded. In other words, $\exists M > 0$ such that $|s_n| \leq M$ for any $n \in \mathbb{N}$.

Examples. The sequence $(n)_{n \in \mathbb{N}}$ is not bounded since the set $\{n : n \in \mathbb{N}\} = \mathbb{N}$ is not bounded. The sequence $((-1)^n)_{n \in \mathbb{N}}$ is bounded since the set $\{(-1)^n : n \in \mathbb{N}\}$ is bounded from above by 1 and bounded from below by -1.

Definition. A sequence is called unbounded if it is not bounded. In other words, $\forall M > 0, \exists n \in \mathbb{N}$ such that $|s_n| > M$.

Proposition. Every convergent sequence is bounded.

Proof. Let $(s_n)_{n \in \mathbb{N}}$ be a convergent sequence. Then there exists some $s \in \mathbb{R}$ such that $(s_n)_{n \in \mathbb{N}} \rightarrow s$. We aim to find a bound for $(s_n)_{n \in \mathbb{N}}$. Let $\epsilon = 1$. There exists some $N \in \mathbb{N}$ such that $|s_n - s| < 1$ for any $n \geq N$. Then for $n \geq N$, we have

$$|s_n| = |(s_n - s) + s| \leq |s_n - s| + |s| < 1 + |s|.$$

Let $M = \max\{|s_1|, |s_2|, \dots, |s_{N-1}|, 1 + |s|\}$. Then

$$|s_n| \leq M, \quad \forall n = 1, \dots, N-1$$

and

$$|s_n| < 1 + |s| \leq M, \quad \forall n \geq N.$$

Hence, $|s_n| \leq M$ for any $n \in \mathbb{N}$ and the sequence $(s_n)_{n \in \mathbb{N}}$ is bounded by definition. $\square$

Theorem. (Limit Laws) Let $(s_n)_{n \in \mathbb{N}}$ and $(t_n)_{n \in \mathbb{N}}$ be two convergent sequences. Let $s = \lim_{n \rightarrow \infty} s_n$ and $t = \lim_{n \rightarrow \infty} t_n$. Then

- (1) $\lim_{n \rightarrow \infty} (s_n \pm t_n) = s \pm t$.
- (2) $\lim_{n \rightarrow \infty} (s_n \pm c) = s \pm c$, for any $c \in \mathbb{R}$.
- (3) $\lim_{n \rightarrow \infty} (s_n t_n) = st$.
- (4) $\lim_{n \rightarrow \infty} (cs_n) = cs$, for any $c \in \mathbb{R}$.
- (5) $\lim_{n \rightarrow \infty} \frac{1}{s_n} = \frac{1}{s}$, provided that $s_n \neq 0$ for any $n \in \mathbb{N}$ and $s \neq 0$.

We postpone the proof to next lecture. First, we look at the consequence of the limit laws.

Example. Prove that

$$a_n = \frac{n^3 + 6n^2 + 7}{4n^3 + 3n + 4} \rightarrow \frac{1}{4}, \quad \text{as } n \rightarrow \infty.$$

Proof. We divide both sides by n^3 to rewrite a_n as

$$a_n = \frac{1 + \frac{6}{n} + \frac{7}{n^3}}{4 + \frac{3}{n^2} + \frac{4}{n^3}}.$$

Denote $b_n = 1 + \frac{6}{n} + \frac{7}{n^3}$ and $c_n = 4 + \frac{3}{n^2} + \frac{4}{n^3}$. Since the sequence $(\frac{1}{n})_{n \in \mathbb{N}}$ converges to 0 (we will use this as a fact), we use (3) and (4) to deduce that the sequences $(\frac{6}{n})_{n \in \mathbb{N}}$ and $(\frac{7}{n^3})_{n \in \mathbb{N}}$ converge to 0. Therefore, by (2) we conclude that $(b_n)_{n \in \mathbb{N}} = (1 + \frac{6}{n} + \frac{7}{n^3})_{n \in \mathbb{N}} \rightarrow 1 + 0 + 0 = 1$. Similarly, we infer that $(c_n)_{n \in \mathbb{N}} \rightarrow 4$. Now, we apply (5) to get

$$\left(\frac{1}{c_n} \right)_{n \in \mathbb{N}} \rightarrow \frac{1}{4},$$

since $c_n \neq 0$ for any $n \in \mathbb{N}$ and $4 \neq 0$. Using (3) we conclude that

$$(a_n)_{n \in \mathbb{N}} = \left(b_n \cdot \frac{1}{c_n} \right)_{n \in \mathbb{N}} \rightarrow \frac{1}{4},$$

as desired. $\square$

9 Lecture 9

Limit theorems for sequences

Theorem. (Limit Laws) Let (s_n) and (t_n) be two convergent sequences. Let $s = \lim_{n \rightarrow \infty} s_n$ and $t = \lim_{n \rightarrow \infty} t_n$. Then

- (1) $\lim_{n \rightarrow \infty} (s_n \pm t_n) = s \pm t$.
- (2) $\lim_{n \rightarrow \infty} (s_n \pm c) = s \pm c$, for any $c \in \mathbb{R}$.
- (3) $\lim_{n \rightarrow \infty} (s_n t_n) = st$.
- (4) $\lim_{n \rightarrow \infty} (cs_n) = cs$, for any $c \in \mathbb{R}$.
- (5) $\lim_{n \rightarrow \infty} \frac{1}{s_n} = \frac{1}{s}$, provided that $s_n \neq 0$ for any $n \in \mathbb{N}$.

Proof. (1) Let $\epsilon > 0$. For $\epsilon/2$, there exists $N_1, N_2 \in \mathbb{N}$ such that

$$|s_n - s| < \frac{\epsilon}{2}, \quad \forall n \geq N_1$$

and

$$|t_n - t| < \frac{\epsilon}{2}, \quad \forall n \geq N_2.$$

Let $N = \max\{N_1, N_2\}$. Then for any $n \geq N$, we have

$$|s_n - s| < \frac{\epsilon}{2} \quad \text{and} \quad |t_n - t| < \frac{\epsilon}{2}$$

which implies that

$$|(s_n + t_n) - (s + t)| \leq |s_n - s| + |t_n - t| < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon.$$

Thus, we have proven the sequence $(s_n + t_n)_{n \in \mathbb{N}} \rightarrow s + t$. A similar proof shows that the sequence $(s_n - t_n)_{n \in \mathbb{N}} \rightarrow s - t$.

(2) Let $c \in \mathbb{R}$. Let $(t_n)_{n \in \mathbb{N}}$ be the constant sequence $\{c, c, c, \dots\}$. $\forall \epsilon > 0$, $\exists N = 1$ such that $|t_n - c| = 0 < \epsilon$, for any $n \geq N$. Hence, $(t_n)_{n \in \mathbb{N}} \rightarrow c$. Using (1), we infer $(s_n \pm c)_{n \in \mathbb{N}} \rightarrow s \pm c$.

(3) Note that

$$|s_n t_n - st| = |s_n t_n - s_n t + s_n t - st| \leq |s_n| |t_n - t| + |s_n - s| |t|.$$

Let $\epsilon > 0$. Since every convergent sequence is bounded, there exists some $M > 0$ such that $|s_n| \leq M$ for all $n \in \mathbb{N}$. For $\frac{\epsilon}{2M} > 0$, there exists some $N_1 \in \mathbb{N}$ such that

$$|t_n - t| < \frac{\epsilon}{2M}, \quad \forall n \geq N_1.$$

For $\frac{\epsilon}{2|t|+1} > 0$, there exists some $N_2 \in \mathbb{N}$ such that

$$|s_n - s| < \frac{\epsilon}{2|t|+1}, \quad \forall n \geq N_2.$$

Let $N = \max\{N_1, N_2\}$. Now, for any $n \geq N$,

$$|s_n t_n - st| \leq |s_n| |t_n - t| + |t| |s_n - s| \leq M \cdot \frac{\epsilon}{2M} + |t| \cdot \frac{\epsilon}{2|t| + 1} = \frac{\epsilon}{2} + \frac{\epsilon}{2} \cdot \frac{|t|}{|t| + \frac{1}{2}} < \epsilon.$$

(4) Let $c \in \mathbb{R}$. Let $(t_n)_{n \in \mathbb{N}}$ be the constant sequence $\{c, c, c, \dots\}$. Since $(t_n)_{n \in \mathbb{N}} \rightarrow c$, from (3) we deduce that $(cs_n)_{n \in \mathbb{N}} \rightarrow cs$.

(5) We refer the proof to the textbook. □

Theorem. (Power theorem) Let (s_n) be a convergent sequence. Let $s = \lim_{n \rightarrow \infty} s_n$. For any $p > 0$, we have $(s_n^p) \rightarrow s^p$.

Definition. The sequence $(s_n)_{n \in \mathbb{N}}$ is said to diverge to ∞ if $\forall M > 0, \exists N \in \mathbb{N}$ such that $\forall n \geq N$, we have $s_n > M$. We denote $\lim_{n \rightarrow \infty} s_n = \infty$.

Definition. The sequence $(s_n)_{n \in \mathbb{N}}$ is said to diverge to $-\infty$ if $\forall M < 0, \exists N \in \mathbb{N}$ such that $\forall n \geq N$, we have $s_n < M$. We denote $\lim_{n \rightarrow \infty} s_n = -\infty$.

Remark. If a sequence (s_n) diverges to ∞ or diverges to $-\infty$, then it's divergent.

Informal discussion. In plain English, the criteria for diverging to ∞ reads: For every (usually large) positive number $M > 0$, one can find a point on the sequence $N \in \mathbb{N}$ such that beyond that point (every term after $n \geq N$), all values of the sequence exceed M .

For example, given $M = 1$, we can find $N_1 \in \mathbb{N}$ such that whenever $n \geq N_1$, we always have $s_n > M = 1$. So s_n exceeds the value of 1 whenever $n \geq N_1$. For the value $M = 10$, we can find $N_{10} \in \mathbb{N}$ (which may be much larger than N_1) such that whenever $n \geq N_{10}$, we always have $s_n > M = 10$. For the value $M = 100$, we can find $N_{100} \in \mathbb{N}$ (which may be much larger than N_{10}) such that whenever $n \geq N_{100}$, we always have $s_n > M = 100$. By continuing like this, we see that the sequence (s_n) is diverging to infinity.

Definition. We say that a sequence (s_n) has a limit or the limit exists if (s_n) converges (to a real number) or diverges to $\pm\infty$.

Example. Prove that $\lim_{n \rightarrow \infty} n^2 = \infty$.

Proof. Let $M > 0$. Let $N \in \mathbb{N}$ such that $N > \sqrt{M}$. Then, for any $n \geq N$,

$$n^2 \geq N^2 > (\sqrt{M})^2 = M.$$

Hence, we conclude that $\lim_{n \rightarrow \infty} n^2 = \infty$. □

10 Lecture 10

Monotone sequences and Cauchy sequences

We are able to verify that a sequence (s_n) converges by guessing a limit s first (we may carry some intuitions from Calculus) and then check the ϵ - N definition holds true for $(s_n) \rightarrow s$. In this section, we present some concepts and theorems which allow us to conclude convergence/divergence without knowing their limit in advance.

Definition. Let $(s_n)_{n \in \mathbb{N}}$ be a sequence. We say that $(s_n)_{n \in \mathbb{N}}$ is an increasing sequence if $\forall n \in \mathbb{N}$ we have $s_n \leq s_{n+1}$. We say that $(s_n)_{n \in \mathbb{N}}$ is a decreasing sequence if $\forall n \in \mathbb{N}$ we have $s_n \geq s_{n+1}$. We say that $(s_n)_{n \in \mathbb{N}}$ is a monotone sequence if it is increasing or decreasing.

Example. The sequence $(1 - \frac{1}{n})_{n \in \mathbb{N}}$ is an increasing sequence. The sequence $((-1)^n)_{n \in \mathbb{N}}$ is not monotone.

Theorem. (Monotone Convergence Theorem) Every bounded and monotone sequence is convergent.

Proof. Let (s_n) be an increasing sequence. Denote $S = \{s_n : n \in \mathbb{N}\}$ which is bounded by assumption. By the completeness axiom, $\sup(S) \in \mathbb{R}$ exists and we denote $s = \sup(S)$. We aim to prove that $(s_n) \rightarrow s$.

Let $\epsilon > 0$. By the proposition in class, there exists some $N \in \mathbb{N}$ such that

$$s - \epsilon < s_N \leq s.$$

Since (s_n) is increasing, for any $n \geq N$ we have $s_n \geq s_N$. Hence, for any $n \geq N$,

$$s - \epsilon < s_N \leq s_n \leq s.$$

It follows that

$$|s_n - s| < \epsilon, \quad \forall n \geq N.$$

Thus, $(s_n) \rightarrow s$.

Let (t_n) be a decreasing sequence. We consider the sequence $(-t_n)$ which is increasing. Since (t_n) is bounded, we infer that $(-t_n)$ is bounded. Hence, using the above argument, we deduce that $(-t_n)$ is convergent. By limit law, the scalar multiplication of a convergent sequence is convergent. Thus, (t_n) is convergent. $\square$

Remark. Equivalent ways of stating MCT: (1) every increasing sequence which is bounded from above is convergent; (2) every decreasing sequence which is bounded from below is convergent.

Example. Consider the sequence $(a_n)_{n \in \mathbb{N}}$ defined by $a_1 = 1$ and $a_{n+1} = \sqrt{3 + 2a_n}$, for any $n \in \mathbb{N}$. Show that $(a_n)_{n \in \mathbb{N}}$ is convergent.

Proof. Let P_n be the statement " $0 \leq a_n \leq 3$ ". The base case P_1 holds by direct verification. Now we assume that P_n holds for some $n \in \mathbb{N}$. By the induction hypothesis, we have $0 \leq a_{n+1} = \sqrt{3 + 2a_n} \leq \sqrt{9} = 3$. Hence, P_{n+1} holds. By induction, $0 \leq a_n \leq 3$ holds for any $n \in \mathbb{N}$. Thus, the sequence $(a_n)_{n \in \mathbb{N}}$ is bounded from above.

Now we compute for any $n \in \mathbb{N}$,

$$\begin{aligned} 0 \leq a_n \leq 3 &\implies a_n^2 - 2a_n - 3 \leq 0 \implies a_n^2 \leq 2a_n + 3 \\ &\implies a_n = |a_n| \leq \sqrt{2a_n + 3} \implies a_n \leq a_{n+1}. \end{aligned}$$

Since $(a_n)_{n \in \mathbb{N}}$ is increasing and bounded from above, by MCT, $(a_n)_{n \in \mathbb{N}}$ is convergent. $\square$

11 Lecture 11

Theorem. Let (s_n) be an unbounded sequence.

- (1) If (s_n) is increasing, then $\lim_{n \rightarrow \infty} s_n = \infty$.
- (2) If (s_n) is decreasing, then $\lim_{n \rightarrow \infty} s_n = -\infty$.

Proof. (1) Since (s_n) is unbounded, it is either unbounded from below or unbounded from above. Since the sequence is increasing, it is bounded from below by s_1 . Thus, it is unbounded from above. By definition, for any $M > 0$, there exists some $N \in \mathbb{N}$ such that $s_N > M$. By monotonicity, for $n \geq N$, we have $s_n \geq s_N > M$. By definition, (s_n) diverges to infinity and $\lim_{n \rightarrow \infty} s_n = \infty$.

(2) The proof is similar to (1). □

Corollary. Every monotone sequence has a limit. Consequently, if (s_n) is monotone, then $\lim_{n \rightarrow \infty} s_n$ exists.

Proof. Let (s_n) be monotone sequence. If (s_n) is bounded, then MCT implies that (s_n) is convergent. If (s_n) is unbounded, then the above theorem implies that (s_n) diverges to $\pm\infty$. □

Definition. Let (s_n) be a sequence. We define

$$\limsup s_n = \lim_{N \rightarrow \infty} v_N = \lim_{N \rightarrow \infty} \sup\{s_n : n > N\}$$

and

$$\liminf s_n = \lim_{N \rightarrow \infty} u_N = \lim_{N \rightarrow \infty} \inf\{s_n : n > N\}.$$

Remark. Recall that for any nonempty subset $S \subseteq \mathbb{R}$,

$$\sup(S) = \begin{cases} \text{a real number,} & \text{if } S \text{ is bounded from above} \\ \infty, & \text{otherwise.} \end{cases}$$

Hence, $v_N = \sup\{s_n : n > N\}$ is a real number or ∞ , for any $N \in \mathbb{N}$. Thus, $(v_N)_{N \in \mathbb{N}}$ may not be a sequence of real numbers. We make an exception only in here that the element of the sequence is allowed to belong to $\mathbb{R} \cup \{\pm\infty\}$. The reason is that we can prove $(v_N)_{N \in \mathbb{N}}$ is decreasing and the above corollary can be applied to $\lim_{N \rightarrow \infty} v_N$. In fact, we have proved that for any nonempty $S \subseteq T \subseteq \mathbb{R}$, we have $\sup(S) \leq \sup(T)$. Therefore,

$$v_{N+1} = \sup(\{s_n : n > N+1\}) \leq \sup(\{s_n : n > N\}) = v_N,$$

for any $N \in \mathbb{N}$. Thus, $(v_N)_{N \in \mathbb{N}}$ is decreasing. If $(v_N)_{N \in \mathbb{N}}$ is a sequence of real numbers, then the above corollary implies that $\lim_{N \rightarrow \infty} v_N$ exists. If $v_M = \infty$ for some $M \in \mathbb{N}$, we will prove in a homework problem that $v_N = \infty$ for all $N \in \mathbb{N}$ and thus $\lim_{N \rightarrow \infty} v_N = \infty$.

Similar consideration also applies to $\liminf s_n$.

Examples.

- (1) Consider the sequence $(s_n)_{n \in \mathbb{N}} = ((-1)^n)_{n \in \mathbb{N}}$. Then for any $N \in \mathbb{N}$,

$$v_N = \sup\{s_n : n > N\} = \sup\{1, -1\} = 1$$

and

$$u_N = \inf\{s_n : n > N\} = \inf\{1, -1\} = -1.$$

Thus, $\limsup s_n = \lim_{N \rightarrow \infty} v_N = 1$ and $\liminf s_n = \lim_{N \rightarrow \infty} u_N = -1$.

(2) Consider the sequence $(t_n)_{n \in \mathbb{N}} = \{1, 2, 3, \dots\}$. Then for any $N \in \mathbb{N}$,

$$v_N = \sup\{t_n : n > N\} = \sup\{N+1, N+2, \dots\} = \infty$$

and

$$u_N = \inf\{t_n : n > N\} = \inf\{N+1, N+2, \dots\} = N+1.$$

Hence, $\limsup t_n = \lim_{N \rightarrow \infty} v_N = \infty$ and $\liminf t_n = \lim_{N \rightarrow \infty} u_N = \infty$.

Theorem. Let $(s_n)_{n \in \mathbb{N}}$ be a sequence.

(1) If $\lim_{n \rightarrow \infty} s_n$ exists (as a real number or $\pm\infty$), then $\liminf s_n = \limsup s_n = \lim_{n \rightarrow \infty} s_n$.

(2) If $\limsup s_n = \liminf s_n$, then $\lim_{n \rightarrow \infty} s_n$ exists and $\liminf s_n = \limsup s_n = \lim_{n \rightarrow \infty} s_n$.

We refer to the textbook for the proof. Theorem part (1) is usually used to conclude divergence, while part (2) is usually used to conclude convergence. See the above two examples.

12 Lecture 12

Definition. A sequence (s_n) is called a Cauchy sequence if for every $\epsilon > 0$ there exists some $N \in \mathbb{N}$ such that $\forall n, m \geq N$, we have $|s_m - s_n| < \epsilon$.

Note. Notice that the definition only refers to elements of the sequence (s_n) and does not mention any limit s (in contrast with the definition of a convergent sequence). We will show eventually that

$$\text{Cauchy sequence} \iff \text{convergent sequence}.$$

Lemma. If (s_n) is convergent, then (s_n) is Cauchy.

Proof. Let $(s_n) \rightarrow s \in \mathbb{R}$. Let $\epsilon > 0$. For $\frac{\epsilon}{2} > 0$, there exists some $N \in \mathbb{N}$ such that

$$|s_n - s| < \frac{\epsilon}{2} \quad \text{for all } n \geq N.$$

Thus, for any $n, m \geq N$, we have

$$|s_n - s_m| = |(s_n - s) + (s - s_m)| \leq |s_n - s| + |s - s_m| < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon.$$

Hence, the sequence (s_n) is Cauchy by definition. $\square$

In order to show Cauchy sequence is convergent, we first show Cauchy sequence is bounded.

Lemma. If (s_n) is Cauchy, then (s_n) is bounded.

Proof. Let $\epsilon = 1$. There exists some $N \in \mathbb{N}$ such that

$$|s_n - s_m| < \epsilon, \quad \forall n, m \geq N.$$

In particular, $|s_n - s_N| < 1$ for all $n \geq N$. Then for any $n \geq N$, we have

$$|s_n| = |(s_n - s_N) + s_N| \leq |s_n - s_N| + |s_N| < 1 + |s_N|.$$

Let $M = \max\{|s_1|, |s_2|, \dots, |s_{N-1}|, 1 + |s_N|\}$. Then $|s_n| \leq M$ for all $n \in \mathbb{N}$. $\square$

Now we are ready to prove the most significant result of the course.

Theorem. Let (s_n) be a sequence of real numbers. Then (s_n) is Cauchy if and only if (s_n) is convergent.

Proof. The first lemma showed the backward implication, so we only focus on the forward implication. Let (s_n) be a Cauchy sequence. Since (s_n) is bounded, for any $N \in \mathbb{N}$, $v_N = \sup\{s_n : n > N\} \in \mathbb{R}$ by completeness axiom. Also, for any $N \in \mathbb{N}$, $v_N = \sup\{s_n : n > N\} \geq s_{N+1} \geq -M$ for some $M > 0$, since (s_n) is bounded. Since $(v_N)_{N \in \mathbb{N}}$ is decreasing and bounded from below, MCT implies $(v_N)_{N \in \mathbb{N}}$ is convergent and thus $\limsup s_n = \lim_{N \rightarrow \infty} v_N$ is a real number. Similarly, $\liminf s_n$ is a real number.

By the theorem from last lecture, it suffices to show that $\limsup s_n = \liminf s_n$. Let $\epsilon > 0$. There exists some $N \in \mathbb{N}$ such that

$$\forall n, m \geq N, \text{ we have } |s_n - s_m| < \epsilon \implies \forall n, m \geq N, \text{ we have } s_n < s_m + \epsilon. \quad (*)$$

Fix some $m \geq N$. Then $s_m + \epsilon$ is an upper bound of $\{s_n : n > N\}$ and $v_N = \sup\{s_n : n > N\} \leq s_m + \epsilon$. This argument works for any $m \geq N$, which implies that $v_N - \epsilon$ is a lower bound of $\{s_m : m > N\}$. Thus, $v_N - \epsilon \leq \inf\{s_m : m > N\} = u_N$. Now, by monotonicity of (u_M) and (v_M) , we get

$$\epsilon + \liminf s_n = \epsilon + \lim_{M \rightarrow \infty} u_M \geq \epsilon + u_N \geq v_N \geq \lim_{M \rightarrow \infty} v_M = \limsup s_n.$$

Since this holds for any $\epsilon > 0$, we have $\liminf s_n \geq \limsup s_n$. It is easy to prove that $\limsup s_n \geq \liminf s_n$. Thus, $\limsup s_n = \liminf s_n$ and (s_n) is convergent. $\square$

13 Lecture 13

Subsequences

Let (s_n) be a sequence. We can construct a new sequence from (s_n) by “extracting” terms. For example, we can extract $s_3, s_7, s_8, s_{10}, s_{100}, s_{125}, \dots$ from the sequence (s_n) and form a new sequence (t_n) :

$$\begin{aligned}t_1 &= s_3 \\t_2 &= s_7 \\t_3 &= s_8 \\t_4 &= s_{10} \\t_5 &= s_{100} \\t_6 &= s_{125} \\&\dots\end{aligned}$$

We refer to (t_n) as a subsequence of (s_n) . To be precise, we state the definition of a subsequence.

Definition. Let $(s_n)_{n \in \mathbb{N}}$ be a sequence. A subsequence of $(s_n)_{n \in \mathbb{N}}$ is a sequence of the form $(s_{n_k})_{k \in \mathbb{N}}$ such that $n(\cdot)$ is a strictly increasing function from $\mathbb{N}$ to $\mathbb{N}$, i.e.,

$$n_1 < n_2 < n_3 < \dots$$

Here we used the convention that $n_k = n(k)$.

Example. Let $s_n = (-1)^n$. We can define a strictly increasing function $n_k = n(k) = 2k$ and thus obtain a subsequence

$$(s_{2k})_{k \in \mathbb{N}} = \{s_2, s_4, s_6, \dots\} = \{1, 1, 1, \dots\}.$$

We can also define another strictly increasing function $n_k = n(k) = 2k + 1$ and the subsequence reads

$$(s_{2k+1})_{k \in \mathbb{N}} = \{s_3, s_5, s_7, \dots\} = \{-1, -1, -1, \dots\}.$$

Note. We emphasize that the subsequence keeps the order from the original sequence. To distinguish the subsequence from the original sequence while still keeping the idea of “subsequence”, we use the notation $(s_{n_k})_{k \in \mathbb{N}}$ which is indexed by $k \in \mathbb{N}$, not $n \in \mathbb{N}$.

Theorem. If $(s_n)_{n \in \mathbb{N}} \rightarrow s$ where $s \in \mathbb{R}$, then every subsequence of $(s_n)_{n \in \mathbb{N}}$ converges to s .

Proof. Let $(s_{n_k})_{k \in \mathbb{N}}$ be a subsequence. Note that $n_k \geq k$, for any $k \in \mathbb{N}$. This is easy to prove by induction and monotonicity of n_k . Let $\epsilon > 0$. There exists some $N \in \mathbb{N}$ such that $|s_n - s| < \epsilon$ for any $n \geq N$. For any $k \geq N$, we have $n_k \geq k \geq N$. Hence,

$$|s_{n_k} - s| < \epsilon, \quad \forall k \geq N.$$

We thus proved that $(s_{n_k})_{k \in \mathbb{N}} \rightarrow s$. □

Theorem. Let $(s_n)_{n \in \mathbb{N}}$ be a sequence and $s \in \mathbb{R}$. If every subsequence of $(s_n)_{n \in \mathbb{N}}$ has a further

subsequence which converges to s , then $(s_n)_{n \in \mathbb{N}} \rightarrow s$.

We aim to prove the theorem by contradiction. Note that we usually prove convergence directly by definition/limit laws/Cauchy sequence. This theorem is one of the few cases where we prove by contradiction.

Proof. For the sake of contradiction, we assume that (s_n) does not converge to s . Then $\exists \epsilon > 0$ such that $\forall N \in \mathbb{N}$, there exists some $n \geq N$ with $|s_n - s| \geq \epsilon$. Let $N = 1$, there exists some $n_1 \geq 1$ such that $|s_{n_1} - s| \geq \epsilon$. Let $N = 1 + n_1$, there exists some $n_2 \geq 1 + n_1$ with $|s_{n_2} - s| \geq \epsilon$. We repeat this process and get a subsequence $(s_{n_k})_{k \in \mathbb{N}}$. The rest of the proof is left as a homework exercise. $\square$

Theorem. (Bolzano-Weierstrass Theorem) Every bounded sequence has a convergent subsequence.

Sketch of the proof. Here we present the classical bisection technique in analysis. Let $(s_n)_{n \in \mathbb{N}}$ be a bounded sequence. There exists some $M > 0$ such that $-M \leq s_n \leq M$, $\forall n \in \mathbb{N}$. We consider the subintervals $[-M, 0]$ and $[0, M]$. Since there are infinitely many elements in the sequence $(s_n)_{n \in \mathbb{N}}$, there is at least one subinterval that contains infinitely many elements from $(s_n)_{n \in \mathbb{N}}$, say $I_1 = [-M, 0]$. We pick some element from $(s_n)_{n \in \mathbb{N}}$ in I_1 and call it s_{n_1} . Next we consider two subintervals $[-M, -M/2]$ and $[-M/2, 0]$ of I_1 . For the same reason, there exists at least one subinterval that contains infinitely many elements from $(s_n)_{n \geq n_1+1}$, say $I_2 = [-M/2, 0]$. We then pick some element from $(s_n)_{n \geq n_1+1}$ in I_2 and call it s_{n_2} . We repeat this process and get a list of intervals $I_1, I_2, I_3, \dots$ and a subsequence $(s_{n_k})_{k \in \mathbb{N}}$ with

$$s_{n_k} \in I_k, \quad \forall k \in \mathbb{N}.$$

Note that the fact that each I_k has infinitely many elements allows us to continue this process forever. Fix $k \in \mathbb{N}$. It is easy to see that for $m \geq k$, we have

$$|s_{n_k} - s_{n_m}| \leq \text{length of } I_k = \frac{2M}{2^k}.$$

It is easy to prove that $(s_{n_k})_{k \in \mathbb{N}}$ is Cauchy by definition and thus convergent. $\square$

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Definition. Let $(s_n)_{n \in \mathbb{N}}$ be a sequence. A subsequential limit is a real number or $\pm\infty$ which is the limit of some subsequence of $(s_n)_{n \in \mathbb{N}}$.

Example. Let $(s_n)_{n \in \mathbb{N}} = ((-1)^n)_{n \in \mathbb{N}}$. We see that 1 is the subsequential limit of (s_n) since the subsequence consisting of even-numbered terms converges to 1. Similarly, -1 is a subsequential limit. The set $S = \{1, -1\}$ is exactly the set of all subsequential limits of (s_n) .

Theorem. Let (s_n) be a sequence and S be the set of all subsequential limits of (s_n) .

- (1) S is nonempty.
- (2) $\sup S = \limsup s_n$ and $\inf S = \liminf s_n$.
- (3) $\lim_{n \rightarrow \infty} s_n$ exists if and only if S has exactly one element.

Proof. (1) *Case 1:* If (s_n) is bounded, then Bolzano-Weierstrass Theorem implies that (s_n) has a convergent subsequence. Thus S is nonempty.

Case 2: If (s_n) is unbounded from above. We aim to construct a subsequence $(s_{n_k})_{k \in \mathbb{N}}$ that diverges to ∞ . Let $M = 1$. Then there exists some $n_1 \in \mathbb{N}$ such that $s_{n_1} > M = 1$. Note that $(s_n)_{n \geq n_1+1}$ is also unbounded from above. (Otherwise, if $(s_n)_{n \geq n_1+1}$ is bounded from above, then $(s_n)_{n \in \mathbb{N}}$ is bounded from above since there are only finitely many terms in $\{s_1, s_2, \dots, s_{n_1}\}$. Contradiction!) Let $M = 2$. There exists some $n_2 \geq n_1 + 1$ such that $s_{n_2} > M = 2$. We repeat this process and get a subsequence $(s_{n_k})_{k \in \mathbb{N}}$ which satisfies $s_{n_k} > k, \forall k \in \mathbb{N}$. It is easy to prove that $(s_{n_k})_{k \in \mathbb{N}}$ diverges to ∞ and thus S contains ∞ .

Case 3: If (s_n) is unbounded from below. The proof is similar and thus we omit it here.

(2) and (3): We omit the proof here. □

Series

A series originates from a sequence. Suppose we are given a sequence $(a_n)_{n \in \mathbb{N}}$ and we compute the sum up to n

$$s_n = \sum_{k=1}^n a_k = a_1 + a_2 + \dots + a_n.$$

We call $(s_n)_{n \in \mathbb{N}}$ the sequence of partial sums associated to (a_n) .

Definition. We say that the series $\sum_{n=1}^{\infty} a_n$ converges if the sequence of partial sums $(s_n)_{n \in \mathbb{N}}$ converges and we write

$$s = \lim_{n \rightarrow \infty} s_n = \sum_{n=1}^{\infty} a_n$$

A series that does not converge is said to diverge. In the cases when a series diverges to $\pm\infty$, we write $\sum_{n=1}^{\infty} a_n = \pm\infty$.

Note. Sometimes we simply write $\sum a_n$, instead of $\sum_{n=1}^{\infty} a_n$, if no confusion can arise. In fact, the starting index doesn't affect convergence or divergence of a series. For example, $\sum_{n=1}^{\infty} a_n$ converges or diverges simultaneously with $\sum_{n=100}^{\infty} a_n$. We will be concerned with properties of a series besides its exact value or what the starting index is.

Examples. Consider the series $\sum_{n=1}^{\infty} \frac{1}{n^2}$. Let $s_n = \sum_{k=1}^n \frac{1}{k^2}$ be the partial sum. Then $(s_n)_{n \in \mathbb{N}}$ is an increasing sequence. For any $n > 1$, we have

$$\begin{aligned} s_n &= \frac{1}{1^2} + \frac{1}{2^2} + \cdots + \frac{1}{n^2} \leq 1 + \frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \cdots + \frac{1}{(n-1) \cdot n} \\ &= 1 + \left(\frac{1}{1} - \frac{1}{2} \right) + \left(\frac{1}{2} - \frac{1}{3} \right) + \cdots + \left(\frac{1}{n-1} - \frac{1}{n} \right) = 2 - \frac{1}{n} \leq 2, \quad \forall n \in \mathbb{N}. \end{aligned}$$

Hence, the sequence (s_n) is bounded from above. By MCT, $(s_n)_{n \in \mathbb{N}}$ is convergent and thus the series $\sum_{n=1}^{\infty} \frac{1}{n^2}$ is convergent by definition.

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Examples.

- (1) The series $\sum_{n=1}^{\infty} \frac{1}{n}$ is divergent. This is left as a homework problem.
- (2) Consider the geometric series $\sum_{n=0}^{\infty} r^n$ ($r \in \mathbb{R}$, $r \neq 1$). Then for $n \geq 0$, we compute the partial sum

$$s_n = 1 + r + \cdots + r^n = \sum_{k=0}^n r^k = \frac{1 - r^{n+1}}{1 - r}.$$

For $|r| < 1$, we have proved in a homework problem that

$$\lim_{n \rightarrow \infty} s_n = \lim_{n \rightarrow \infty} \frac{1 - r^{n+1}}{1 - r} = \frac{1}{1 - r}.$$

Therefore, the sequence of partial sums $(s_n)_{n \geq 0}$ is convergent and the series $\sum_{n=0}^{\infty} r^n$ is convergent by definition.

Definition. A series $\sum a_n$ is said to be absolutely convergent if $\sum |a_n|$ converges.

Example. Consider the geometric series $\sum_{n=0}^{\infty} r^n$ ($r \in \mathbb{R}$, $r \neq 1$). A similar proof as in above shows that $\sum_{n=0}^{\infty} r^n$ is absolutely convergent if $|r| < 1$.

Definition. We say that a series $\sum a_n$ satisfies the Cauchy criterion if the sequence of partial sums (s_n) is a Cauchy sequence, i.e., $\forall \epsilon > 0, \exists N \in \mathbb{N}$ such that $|s_n - s_m| < \epsilon$, for any $n, m \geq N$.

Remark. Without loss of generality, we may assume that $n \geq m$. Moreover, the case $n = m$ is always satisfied since $|s_n - s_m| = 0 < \epsilon$. Hence, we may modify the definition a little by checking:

$$\forall \epsilon > 0, \exists N \in \mathbb{N} \text{ such that } |s_n - s_{m-1}| < \epsilon, \text{ for any } n \geq m \geq N,$$

which is equivalent to

$$\forall \epsilon > 0, \exists N \in \mathbb{N} \text{ such that } \left| \sum_{k=m}^n a_k \right| < \epsilon, \text{ for any } n \geq m \geq N, \quad (*)$$

Proposition. The series $\sum a_n$ is convergent if and only if it satisfies the Cauchy criterion.

Proof. We have the following equivalence relation:

$$\begin{aligned} \sum a_n \text{ satisfies the Cauchy criterion} &\stackrel{\text{by definition}}{\iff} \text{the sequence of partial sums } (s_n) \text{ is Cauchy} \\ &\iff \text{the sequence of partial sums } (s_n) \text{ is convergent} \stackrel{\text{by definition}}{\iff} \sum a_n \text{ is convergent.} \end{aligned}$$

Hence, the proposition is proved. $\square$

Corollary. (Divergence test) If a series $\sum a_n$ converges, then $\lim_{n \rightarrow \infty} a_n = 0$.

Proof. Since the series $\sum a_n$ converges, it satisfies the Cauchy criterion. In particular, $(*)$ holds for $n = m$, i.e., $\forall \epsilon > 0, \exists N \in \mathbb{N}$ such that $|a_n| < \epsilon$, for any $n \geq N$. Hence, $(a_n) \rightarrow 0$. $\square$

Remark. The contrapositive of the corollary is: If (a_n) is divergent or $\lim a_n \neq 0$, then the series $\sum a_n$ is divergent.

Remark. A common misconception about series is that $\lim_{n \rightarrow \infty} a_n = 0 \implies \sum a_n$ converges. This is false in general! For example, we have $\lim_{n \rightarrow \infty} \frac{1}{n} = 0$ but $\sum \frac{1}{n}$ diverges.

Comparison test. Let $\sum a_n$ be a series where $a_n \geq 0$ for all $n \in \mathbb{N}$.

(1) If $\sum a_n$ converges and $|b_n| \leq a_n, \forall n \in \mathbb{N}$, then $\sum b_n$ converges.

(2) If $\sum a_n = \infty$ and $b_n \geq a_n, \forall n \in \mathbb{N}$, then $\sum b_n = \infty$.

Proof. (1) For any $m, n \in \mathbb{N}$ with $n \geq m$, we have

$$\left| \sum_{k=m}^n b_k \right| \leq \sum_{k=m}^n |b_k| \leq \sum_{k=m}^n a_k.$$

Since the series $\sum a_n$ is convergent, it satisfies the Cauchy criterion. It follows from (*) that $\sum b_n$ also satisfies the Cauchy criterion and thus convergent.

(2) Let (s_n) and (t_n) be the sequences of partial sums for $\sum a_n$ and $\sum b_n$, respectively. Since $b_n \geq a_n$ for all $n \in \mathbb{N}$, we get $t_n \geq s_n$ for all $n \in \mathbb{N}$. The sequence (s_n) is increasing since $a_n \geq 0$ for any $n \in \mathbb{N}$. The sequence (s_n) must be unbounded from above. Otherwise by MCT it implies that (s_n) is convergent and $\sum a_n$ is convergent, contradiction! The sequence (s_n) is unbounded from above and increasing, we get (s_n) diverges to infinity. Since $s_n \leq t_n$ for any $n \in \mathbb{N}$, we conclude that $\lim t_n = \infty$ and thus $\sum b_n = \infty$. $\square$

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Corollary. Absolutely convergent series are convergent.

Proof. Let $\sum a_n$ be absolutely convergent. By definition, $\sum |a_n|$ is convergent. Clearly, we have $|a_n| \leq |a_n|$ for all $n \in \mathbb{N}$. By comparison test, we get $\sum a_n$ converges. $\square$

Comparing a given series with a convergent geometric series using the comparison test is very useful. It is so useful, in fact, that it has been codified into a test of its own: the root test.

Root test. Suppose (a_n) is a sequence and $\limsup |a_n|^{1/n} = \alpha$.

(1) $\alpha < 1 \implies \sum a_n$ converges absolutely.

(2) $\alpha > 1 \implies \sum a_n$ diverges.

(3) $\alpha = 1$ gives no information.

Proof. (1) Since $\alpha < 1$, there exists some $\epsilon > 0$ such that $\alpha + \epsilon < 1$. By definition,

$$\alpha = \limsup |a_n|^{1/n} = \lim_{N \rightarrow \infty} v_N = \lim_{N \rightarrow \infty} (\sup\{|a_n|^{1/n} : n > N\}).$$

There exists some $N \in \mathbb{N}$ such that

$$v_N = \sup\{|a_n|^{1/n} : n > N\} < \alpha + \epsilon.$$

This implies that whenever $n > N$, we have $|a_n|^{1/n} < \alpha + \epsilon$. Raising this inequality to n th power,

$$|a_n| < (\alpha + \epsilon)^n, \quad \forall n \geq N + 1.$$

Since $0 < \alpha + \epsilon < 1$, the geometric series $\sum_{n=N+1}^{\infty} (\alpha + \epsilon)^n$ converges and the comparison test shows that $\sum_{n=N+1}^{\infty} a_n$ converges absolutely. Hence, $\sum_{n=1}^{\infty} a_n$ converges absolutely.

(2) By definition,

$$\alpha = \limsup |a_n|^{1/n} = \lim_{N \rightarrow \infty} v_N = \lim_{N \rightarrow \infty} (\sup\{|a_n|^{1/n} : n > N\}).$$

Since (v_N) is a decreasing sequence, we get

$$v_N = \sup\{|a_n|^{1/n} : n > N\} \geq \alpha, \quad \forall N \in \mathbb{N}.$$

This means that there are infinitely many $n \in \mathbb{N}$ such that $|a_n|^{1/n} \geq 1$. Hence, there are infinitely many $n \in \mathbb{N}$ such that $|a_n| \geq 1$. This implies that (a_n) diverges or $\lim_{n \rightarrow \infty} a_n \neq 0$. By divergence test, $\sum a_n$ is divergent.

(3) For the series $\sum \frac{1}{n}$ and $\sum \frac{1}{n^2}$, the value of α are both equal to 1 (by using elementary calculus). However, $\sum \frac{1}{n}$ is divergent and $\sum \frac{1}{n^2}$ is convergent. $\square$

Ratio test. Let (a_n) be a sequence such that $a_n \geq 0$ for all $n \in \mathbb{N}$.

(1) $\limsup \left| \frac{a_{n+1}}{a_n} \right| < 1 \implies \sum a_n$ converges absolutely.

(2) $\liminf \left| \frac{a_{n+1}}{a_n} \right| > 1 \implies \sum a_n$ diverges.

(3) $\limsup \left| \frac{a_{n+1}}{a_n} \right| \geq 1 \geq \liminf \left| \frac{a_{n+1}}{a_n} \right|$ gives no information.

Proof. Following the notation of the root test, we set $\alpha = \limsup |a_n|^{1/n}$. We use a result from section 12 (which we skipped)

$$\liminf \left| \frac{a_{n+1}}{a_n} \right| \leq \liminf |a_n|^{\frac{1}{n}} \leq \limsup |a_n|^{\frac{1}{n}} \leq \limsup \left| \frac{a_{n+1}}{a_n} \right|.$$

- (1) If $\limsup \left| \frac{a_{n+1}}{a_n} \right| < 1$, then $\alpha < 1$, so the root test implies that $\sum a_n$ converges absolutely.
- (2) If $\liminf \left| \frac{a_{n+1}}{a_n} \right| > 1$, then $\alpha > 1$, so the root test implies that $\sum a_n$ diverges.
- (3) Consider again $\sum \frac{1}{n}$ and $\sum \frac{1}{n^2}$. □

Alternating series and integral tests

We assume familiarity with integral from calculus. Although we will rigorously build up Riemann integration theory later in this course, we first accept that the integral of a function equals the total area under the curve.

Let $\sum a_n$ be a series where the terms a_n are nonnegative. **Integral test** is a principle that we think of the partial sum $\sum_{k=1}^n a_k$ as a sum of areas of rectangles. More precisely, $\sum_{k=1}^n a_k$ represents the total area of the rectangles, each of which has horizontal length 1 and the vertical length a_k . By choosing the function $f(x)$ cleverly, we can compare the partial sum $\sum_{k=1}^n a_k$ with an integral of $f(x)$.

P-test. Let $p > 0$. The series $\sum \frac{1}{n^p}$ converges if and only if $p > 1$.

Proof. If $p > 1$, we compute the partial sum

$$s_n = \sum_{k=1}^n \frac{1}{k^p} \leq 1 + \int_1^n \frac{1}{x^p} dx = 1 + \frac{1}{p-1} \left(1 - \frac{1}{n^{p-1}} \right) < 1 + \frac{1}{p-1} = \frac{p}{p-1}.$$

Hence, by MCT, the sequence (s_n) is convergent and thus $\sum a_n$ converges.

If $0 < p \leq 1$, then

$$\frac{1}{n} \leq \frac{1}{n^p}, \quad \forall n \in \mathbb{N}.$$

Since $\sum \frac{1}{n}$ diverges, we have $\sum \frac{1}{n^p}$ diverges by comparison test. □

Alternating series test. Suppose $(a_n)_{n \in \mathbb{N}}$ is a decreasing sequence of non-negative terms such that $\lim a_n = 0$. Then the series $\sum (-1)^{n+1} a_n$ converges.

We refer the proof to the textbook.

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Continuous functions

We will be studying functions with domain and codomain in $\mathbb{R}$. More precisely, we will focus on functions f such that $\text{dom}(f) \subseteq \mathbb{R}$ and for every $x \in \text{dom}(f)$, we have $f(x) \in \mathbb{R}$. The symbol f represents the function while $f(x)$ represents the value of the function at the point x .

We may have learned the idea of continuity in calculus. Let $x_0 \in \text{dom}(f)$. We ask that $f(x)$ is arbitrarily close to $f(x_0)$ whenever x is sufficiently close to x_0 . We quantify this relation using the following definition.

Definition 1. (ϵ - δ continuity) We say that $f(x)$ is continuous at $x_0 \in \text{dom}(f)$ if

$$\forall \epsilon > 0, \exists \delta > 0 \text{ such that } |x - x_0| < \delta \implies |f(x) - f(x_0)| < \epsilon. \quad (*)$$

We say that $f(x)$ is discontinuous at x_0 if $(*)$ fails. If $(*)$ holds for all $x_0 \in S \subseteq \text{dom}(f)$, then we say f is continuous on S .

Informal discussion. The ϵ - δ definition translates to: for any (small) $\epsilon > 0$, there exists $\delta > 0$ (usually depends on ϵ proportionally and also depends on x_0), such that whenever x is at a distance of at most δ from x_0 , then the distance between $f(x)$ and $f(x_0)$ is at most ϵ . This definition in some sense is similar to the ϵ - N definition of convergence of a sequence.

Example. Prove that the function $f(x) = x$ is continuous on $\mathbb{R}$.

Proof. Let $x_0 \in \mathbb{R}$. Let $\epsilon > 0$. We take $\delta = \epsilon > 0$. Then

$$|x - x_0| < \delta \implies |f(x) - f(x_0)| = |x - x_0| < \delta = \epsilon.$$

Thus, f is continuous at x_0 . Since $x_0 \in \mathbb{R}$ is arbitrary, the function f is continuous on $\mathbb{R}$. □

Remark. More generally, let $c \in \mathbb{R}$, we can show that $f(x) = x + c$ and $g(x) = cx$ are continuous at $x_0 \in \mathbb{R}$ using similar proof.

Example. Prove that $f(x) = x^2$ is continuous on $\mathbb{R}$.

Proof. Let $x_0 \in \mathbb{R}$. Let $\epsilon > 0$. Let $\delta = \min\{1, \epsilon/(1 + 2|x_0|)\} > 0$, then

$$|x - x_0| < \delta \implies |f(x) - f(x_0)| = |x - x_0||x + x_0| \leq \delta(|x - x_0| + 2|x_0|) < \delta(1 + 2|x_0|) \leq \epsilon.$$

Hence, f is continuous at x_0 . Since $x_0 \in \mathbb{R}$ is arbitrary, we conclude that f is continuous on $\mathbb{R}$. □

Definition 2. (sequential continuity) We say that $f(x)$ is sequentially continuous at $x_0 \in \text{dom}(f)$ if

$$\text{for any sequence } (x_n)_{n \in \mathbb{N}} \rightarrow x_0, \text{ we have } (f(x_n))_{n \in \mathbb{N}} \rightarrow f(x_0). \quad (**)$$

Proposition. Definition 1 is equivalent to definition 2.

Note. Since two definitions of continuity are equivalent, we may choose the one that makes the proof easier.

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Example. Suppose that f is continuous at $x_0 \in \text{dom}(f)$, then $|f|$ is continuous at x_0 .

Proof. Note that by reverse triangle inequality, we have

$$||f(x)| - |f(x_0)|| \leq |f(x) - f(x_0)|.$$

Let $\epsilon > 0$. By continuity at x_0 , we have

$$\forall \epsilon > 0, \exists \delta > 0 \text{ such that } |x - x_0| < \delta \implies |f(x) - f(x_0)| < \epsilon.$$

Hence, for this very $\delta > 0$, we have

$$|x - x_0| < \delta \implies ||f(x)| - |f(x_0)|| \leq |f(x) - f(x_0)| < \epsilon.$$

Therefore, $|f|$ is continuous at x_0 by definition. □

Theorem. Let f and g be functions that are continuous at x_0 . Then

- (1) $f \pm g$ is continuous at x_0 .
- (2) $f \cdot g$ is continuous at x_0 .
- (3) $\frac{f}{g}$ is continuous at x_0 , provided that $g(x) \neq 0$.
- (4) cf is continuous at x_0 , for any $c \in \mathbb{R}$.

We prove sequential continuity using limit laws of sequences. Details are omitted.

Example. Let f and g be continuous at $x_0 \in \mathbb{R}$. Prove that $\max\{f, g\}$ is continuous at $x_0 \in \mathbb{R}$.

Proof. Note that

$$\max\{f(x), g(x)\} = \frac{1}{2}(f(x) + g(x)) + \frac{1}{2}|f(x) - g(x)|.$$

Since f and g are continuous at x_0 , we get $(f + g)/2$ and $|f - g|/2$ are continuous at x by the above example and theorem. Hence, $\max\{f, g\}$ is continuous at x_0 . □

Theorem. If $f(x)$ is continuous at x_0 and $g(x)$ is continuous at $f(x_0)$. Then the composition $g \circ f = g(f(\cdot))$ is continuous at x_0 .

Proof. Let $(x_n) \rightarrow x_0$. By sequential continuity of f at x_0 , we have $(f(x_n)) \rightarrow f(x_0)$. By sequential continuity of g at $f(x_0)$, we have $(g(f(x_n))) \rightarrow g(f(x_0))$. Therefore, $g \circ f$ satisfies sequential continuity definition and thus continuous at x_0 . □

Properties of continuous functions

Definition. Let f be a function and $S \subseteq \text{dom}(f)$. We say that f is bounded on S if the set $\{f(x) : x \in S\}$ is bounded, i.e.,

$$\exists M > 0 \text{ such that } |f(x)| \leq M, \forall x \in S.$$

Theorem.(Extreme Value Theorem) Let f be continuous function defined on a closed interval $[a, b]$. Then f is bounded. Moreover, f attains its maximum and minimum values on $[a, b]$, i.e., there exists $x_m, x_M \in [a, b]$ such that

$$f(x_m) \leq f(x) \leq f(x_M), \quad \forall x \in [a, b].$$

We shall only prove the first statement that f is bounded and refer the rest to the textbook.

Proof. For the sake of contradiction, we assume that f is unbounded. Hence, $\forall M > 0, \exists x \in [a, b]$ such that $|f(x)| > M$. Let $M = n$, where $n \in \mathbb{N}$, there exists some $x_n \in [a, b]$ such that $|f(x_n)| > n$. By construction, the sequence (x_n) is bounded. By Bolzano-Weierstrass theorem, there exists a convergent subsequence $(x_{n_k})_{k \in \mathbb{N}}$ and $(x_{n_k})_{k \in \mathbb{N}} \rightarrow x_0$. Moreover, the limit $x_0 \in [a, b]$ (buy this. Can be an exercise if you don't believe it). Since f is continuous at x_0 , we have $f(x_{n_k}) \rightarrow f(x_0)$ and thus $|f(x_{n_k})| \rightarrow |f(x_0)|$. However, since $|f(x_{n_k})| > n_k > k, \forall k \in \mathbb{N}$, the sequence $(|f(x_{n_k})|)$ is unbounded and thus can't be convergent. Contradiction. $\square$

Theorem.(Intermediate Value Theorem) Let I be an interval with points $a, b \in I$ with $a < b$. Let $f : I \rightarrow \mathbb{R}$ be a continuous function. Assume that $y \in \mathbb{R}$ is a point lying between $f(a)$ and $f(b)$ (meaning $f(a) \leq y \leq f(b)$ or $f(b) \leq y \leq f(a)$). Then, there exists at least one $x \in [a, b]$ such that $f(x) = y$.

We only sketch the key points in the proof.

- We assume that $f(a) \leq y \leq f(b)$ and the case $f(b) \leq y \leq f(a)$ is similar. We consider only $f(a) < y < f(b)$, since if $y = f(a)$ or $y = f(b)$, then we are finished.
- The set $S = \{x \in [a, b] : f(x) < y\}$ is nonempty since $a \in S$. The set S is bounded from above by b .
- Completeness axiom implies that $x_0 := \sup(S)$ exists as a real number. Prove that $x_0 \in [a, b]$.
- Prove that $f(x_0) \leq y$ and $f(x_0) \geq y$ by constructing two sequences $(s_n) \rightarrow x_0$ and $(t_n) \rightarrow x_0$ and using sequential continuity.
- Conclude that $f(x_0) = y$.

19 Lecture 19

Uniform continuity

Recall in the criteria of ϵ - δ continuity at $x_0 \in \text{dom}(f)$, the value of $\delta > 0$ usually depends on x_0 and ϵ . For instance, when we prove that $f(x) = x^2$ is continuous at $x_0 \in \mathbb{R}$, we take $\delta = \min\{1, \epsilon/(1+2|x_0|)\}$.

Definition. Let $f(x)$ be defined on $S \subseteq \text{dom}(f)$. We say that $f(x)$ is uniformly continuous on S if

$$\forall \epsilon > 0, \exists \delta > 0, \text{ such that for any } x, y \in S \text{ with } |x - y| < \delta \text{ we have } |f(x) - f(y)| < \epsilon. \quad (*)$$

Note. (1) x and y are both allowed to vary, which is different from ϵ - δ continuity at x_0 where x_0 is fixed and x is closed to x_0 .

(2) The value of δ is independent of the variables x and y .

Remark. A function f is uniformly continuous on $S \subseteq \text{dom}(f)$ implies that f is continuous on S . In general, the backward implication is false.

Example. Prove that $f(x) = \frac{1}{x^2}$ is uniformly continuous on $(1, \infty)$.

Proof. We aim to prove (*). Let $\epsilon > 0$. We take $\delta = \epsilon/2 > 0$. Then whenever $|x - y| < \delta$, we have

$$\begin{aligned} |f(x) - f(y)| &= \left| \frac{1}{x^2} - \frac{1}{y^2} \right| = \left| \frac{(x-y)(x+y)}{x^2y^2} \right| < \delta \cdot \frac{x+y}{x^2y^2} \\ &= \delta \left(\frac{1}{xy^2} + \frac{1}{x^2y} \right) < \delta \cdot 2 = \epsilon. \end{aligned}$$

Hence, f is uniformly continuous on $(1, \infty)$. □

Example. Prove that $f(x) = \frac{1}{x}$ is not uniformly continuous on $(0, \infty)$.

Proof. For the sake of contradiction, we assume that f is uniformly continuous on $(0, \infty)$. For $\epsilon = 1$, there exists $\delta > 0$ such that $|x - y| < \delta$ implies that $|\frac{1}{x} - \frac{1}{y}| < 1$. In particular, for $x \in (0, \infty)$ with $x^2 + \frac{\delta x}{2} - \frac{\delta}{4} < 0$ and $y = x + \frac{\delta}{2}$, we have $|x - y| = \frac{\delta}{2} < \delta$ and

$$\left| \frac{1}{x} - \frac{1}{y} \right| = \left| \frac{1}{x} - \frac{1}{x + \frac{\delta}{2}} \right| = \frac{\frac{\delta}{2}}{x(x + \frac{\delta}{2})} = \frac{\frac{\delta}{2}}{x^2 + \frac{\delta}{2}x} > \frac{\frac{\delta}{2}}{\frac{\delta}{4}} = 2 > 1.$$

Hence, we get a contradiction. □

Theorem. If f is continuous on a closed interval $[a, b]$, then f is uniformly continuous on $[a, b]$.

Proof. For the sake of contradiction, we assume that f is not uniformly continuous on $[a, b]$. Then there exists $\epsilon > 0$ such that for all $\delta > 0$, there exist some $x, y \in [a, b]$ such that $|x - y| < \delta$ and $|f(x) - f(y)| \geq \epsilon$. Let $\delta = \frac{1}{n}$, where $n \in \mathbb{N}$, there exist $x_n, y_n \in [a, b]$ such that $|x_n - y_n| < \frac{1}{n}$ and $|f(x_n) - f(y_n)| \geq \epsilon$. By the Bolzano-Weierstrass Theorem, a subsequence (x_{n_k}) of (x_n) converges and $(x_{n_k})_{k \in \mathbb{N}} \rightarrow x_0 \in [a, b]$. We also have $(y_{n_k})_{k \in \mathbb{N}} \rightarrow x_0$. Since f is continuous at x_0 , we have

$$f(x_0) = \lim_{k \rightarrow \infty} f(x_{n_k}) = \lim_{k \rightarrow \infty} f(y_{n_k}),$$

and thus

$$\lim_{k \rightarrow \infty} [f(x_{n_k}) - f(y_{n_k})] = 0.$$

Since $|f(x_{n_k}) - f(y_{n_k})| \geq \epsilon$ for all k , we get a contradiction. Hence, we conclude that f is uniformly continuous on $[a, b]$. $\square$

20 Lecture 20

Limits of functions

Definition. Let f be defined on (a, b) and $c \in (a, b)$. We say that $L \in \mathbb{R}$ is the limit of f as x approaches c if

$$\forall \epsilon > 0, \exists \delta > 0 \text{ such that } 0 < |x - c| < \delta \implies |f(x) - L| < \epsilon. \quad (*)$$

In this case, we write

$$L = \lim_{x \rightarrow c} f(x).$$

Note. (1) The limit is a real number such that the function gets arbitrarily close to when x is approaching c . It is easy to check that if the limit exists, then it is unique.

(2) The limit of a function makes sense even when the function is not defined at the point. In fact, oftentimes we will be asked to find the limit of a function which is not defined at c .

(3) Recall that ϵ - δ continuity is equivalent to sequential continuity. Similarly, (*) is equivalent to

$$\text{for any sequence } (x_n) \rightarrow c \text{ with } x_n \neq c \forall n \in \mathbb{N}, \text{ we have } (f(x_n)) \rightarrow L.$$

Either one of these two conditions can be used as definition.

Definition. We say that $L \in \mathbb{R}$ is the right limit of f as x approaches c if

$$\forall \epsilon > 0, \exists \delta > 0 \text{ such that } c < x < c + \delta \implies |f(x) - L| < \epsilon. \quad (**)$$

In this case, we write

$$L = \lim_{x \rightarrow c^+} f(x).$$

We have a similar definition for the left limit of f and we write $L = \lim_{x \rightarrow c^-} f(x)$.

Proposition. $\lim_{x \rightarrow c} f(x) = L$ if and only if $\lim_{x \rightarrow c^+} f(x) = \lim_{x \rightarrow c^-} f(x) = L$.

Theorem. (Limit law for functions) Let f_1 and f_2 be two functions for which $L_1 = \lim_{x \rightarrow c} f_1(x)$ and $L_2 = \lim_{x \rightarrow c} f_2(x)$. Then

- (1) $\lim_{x \rightarrow c} a f_1(x) = a \lim_{x \rightarrow c} f_1(x) = a L_1$, for any $a \in \mathbb{R}$.
- (2) $\lim_{x \rightarrow c} f_1(x) \pm f_2(x) = \lim_{x \rightarrow c} f_1(x) \pm \lim_{x \rightarrow c} f_2(x) = L_1 \pm L_2$.
- (3) $\lim_{x \rightarrow c} f_1(x) \cdot f_2(x) = \lim_{x \rightarrow c} f_1(x) \cdot \lim_{x \rightarrow c} f_2(x) = L_1 \cdot L_2$.
- (4) $\lim_{x \rightarrow x_0} \frac{f_1(x)}{f_2(x)} = \frac{\lim_{x \rightarrow x_0} f_1(x)}{\lim_{x \rightarrow x_0} f_2(x)} = \frac{L_1}{L_2}$, provided that $L_2 \neq 0$.

The following proposition provides an familiar way to find the limit of a function.

Proposition. Let f be a function defined on an open interval containing c . Then f is continuous at c if and only if $f(c) = \lim_{x \rightarrow c} f(x)$.

Proof. For the forward implication, it suffices to prove that (*) holds for $L = f(c)$, which follows from the definition of continuity at c . For the backward implication, we use (*) and $|f(c) - f(c)| = 0 < \epsilon$. □

Basic properties of the derivatives

Definition. Let f be a function defined on (x_1, x_2) and $a \in (x_1, x_2)$. We say that f is differentiable at a if

$$\lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$$

exists and is finite. In this case, we write

$$f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}.$$

If f is differentiable at any points $x \in (x_1, x_2)$, we say that f is differentiable on (x_1, x_2) .

Note. It is an easy exercise (by defining $y_n = x_n - a$) that

$$\lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a} = \lim_{h \rightarrow 0} \frac{f(a + h) - f(a)}{h}.$$

Example. Find the derivative of $f(x) = 1$ at $a \in \mathbb{R}$.

Proof. We compute

$$\lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a} = \lim_{x \rightarrow a} \frac{1 - 1}{x - a} = \lim_{x \rightarrow a} \frac{0}{x - a} = 0.$$

Hence, f is differentiable at a and $f'(a) = 0$. □

Example. Find the derivative of $f(x) = x^2$ at any $a \in \mathbb{R}$.

Proof. We compute

$$\lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a} = \lim_{x \rightarrow a} \frac{x^2 - a^2}{x - a} = \lim_{x \rightarrow a} (x + a) = 2a.$$

Hence, $f'(a) = 2a$. □

21 Lecture 21

Basic properties of the derivative

Example. Consider $f(x) = \sqrt{x}$. Is $f(x)$ differentiable at $x = 1$? If so, find its derivative. Is $f(x)$ differentiable at $x = 0$? If so, find its derivative.

Proof. We compute

$$\lim_{x \rightarrow 1} \frac{f(x) - f(1)}{x - 1} = \lim_{x \rightarrow 1} \frac{\sqrt{x} - 1}{x - 1} = \lim_{x \rightarrow 1} \frac{\sqrt{x} - 1}{(\sqrt{x} + 1)(\sqrt{x} - 1)} = \lim_{x \rightarrow 1} \frac{1}{\sqrt{x} + 1} = \frac{1}{2}.$$

Hence, $f'(1) = \frac{1}{2}$. Next, we compute

$$\lim_{x \rightarrow 0^+} \frac{f(x) - f(0)}{x - 0} = \lim_{x \rightarrow 0^+} \frac{\sqrt{x}}{x} = \lim_{x \rightarrow 0^+} \frac{1}{\sqrt{x}}.$$

The function $1/\sqrt{x}$ diverges to infinity as $x \rightarrow 0^+$. Hence, the function $f(x) = \sqrt{x}$ is not differentiable at $x = 0$. $\square$

Example. Let $f(x) = x^n$, where $n \in \mathbb{N}$. Then $f'(x) = nx^{n-1}$ for all $x \in \mathbb{R}$. This is left as an exercise.

Theorem. If f is differentiable at a , then f is continuous at a .

Proof. For $x \neq a$, we have

$$f(x) = (x - a) \frac{f(x) - f(a)}{x - a} + f(a).$$

Hence,

$$\begin{aligned} \lim_{x \rightarrow a} f(x) &= \lim_{x \rightarrow a} \left((x - a) \frac{f(x) - f(a)}{x - a} + f(a) \right) \\ &= \lim_{x \rightarrow a} (x - a) \frac{f(x) - f(a)}{x - a} + \lim_{x \rightarrow a} f(a) = 0 + f(a) = f(a). \end{aligned}$$

Hence, f is continuous at a . $\square$

Remark. The backward implication is false in general. For example, we consider $f(x) = |x|$ at $x = 0$. The function is continuous at $x = 0$. However, we compute

$$\lim_{x \rightarrow 0^+} \frac{f(x) - f(0)}{x - 0} = \lim_{x \rightarrow 0^+} \frac{x}{x} = 1$$

and

$$\lim_{x \rightarrow 0^-} \frac{f(x) - f(0)}{x - 0} = \lim_{x \rightarrow 0^-} \frac{-x}{x} = -1.$$

Since the left limit is not equal to the right limit, the limit doesn't exist and thus the function is not differentiable.

Theorem. Let f and g be differentiable at a . Then

$$(1) \quad (cf)'(a) = cf'(a), \text{ for any } c \in \mathbb{R}.$$

-
- (2) $(f \pm g)'(a) = f'(a) \pm g'(a)$.
- (3) $(fg)'(a) = f'(a)g(a) + f(a)g'(a)$.
- (4) $\left(\frac{f}{g}\right)'(a) = \frac{f'(a)g(a) - f(a)g'(a)}{g^2(a)}$, provided that $g(a) \neq 0$.

Proof. (1) We compute

$$\lim_{x \rightarrow a} \frac{cf(x) - cf(a)}{x - a} = c \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a} = cf'(a).$$

(2) We compute

$$\lim_{x \rightarrow a} \frac{(f + g)(x) - (f + g)(a)}{x - a} = \lim_{x \rightarrow a} \left(\frac{f(x) - f(a)}{x - a} + \frac{g(x) - g(a)}{x - a} \right) = f'(a) + g'(a).$$

Similar proof also works for $(f - g)'(a)$.

(3) We compute

$$\begin{aligned} \lim_{x \rightarrow a} \frac{f(x)g(x) - f(a)g(a)}{x - a} &= \lim_{x \rightarrow a} \left(\frac{(f(x)g(x) - f(x)g(a)) + (f(x)g(a) - f(a)g(a))}{x - a} \right) \\ &= \lim_{x \rightarrow a} \left(f(x) \cdot \frac{g(x) - g(a)}{x - a} + g(a) \cdot \frac{f(x) - f(a)}{x - a} \right) \\ &= f(a)g'(a) + g(a)f'(a). \end{aligned}$$

(4) The proof uses similar algebraic manipulation as in (3) and we omit it here. $\square$

Theorem. (Chain rule) Suppose f is differentiable at a and g is differentiable at $f(a)$, then the composition $g \circ f$ is differentiable at a and $(g \circ f)'(a) = g'(f(a)) \cdot f'(a)$.

We outline the key points in the proof.

- (1) We aim to prove it “sequentially”. Let $(x_n) \rightarrow a$ be a sequence with $x_n \neq a, \forall n \in \mathbb{N}$. We aim to prove

$$\lim_{n \rightarrow \infty} \frac{g(f(x_n)) - g(f(a))}{x_n - a} = g'(f(a)) \cdot f'(a).$$

- (2) Denote $y_n = f(x_n)$. Then we have

$$\frac{g(f(x_n)) - g(f(a))}{x_n - a} = \frac{g(y_n) - g(f(a))}{y_n - f(a)} \cdot \frac{f(x_n) - f(a)}{x_n - a}.$$

- (3) If $y_n \neq f(a)$ beyond some point in the sequence, then we take the limit and conclude the proof. If $y_n = f(a)$ infinitely often, then we prove that

- (a) $f'(a) = 0$ (easy)
- (b) $g \circ f$ is differentiable at a (difficult)
- (c) $(g \circ f)'(a) = 0$ (easy)
- (d) $(g \circ f)'(a) = 0 = g'(f(a)) \cdot f'(a)$.

22 Lecture 22

Mean value theorem

In previous calculus classes, we have seen how the derivative can be used to determine extreme values by finding the critical points. In this section, we prove such results.

Theorem. (finding the critical points) Let f be defined on (a, b) and $x_0 \in (a, b)$. Suppose that f is differentiable at x_0 and $f(x_0)$ is the maximum or minimum over (a, b) , i.e.,

$$f(x_0) \geq f(x), \quad \forall x \in (a, b) \quad \text{or} \quad f(x_0) \leq f(x), \quad \forall x \in (a, b).$$

Then $f'(x_0) = 0$.

Proof. For the sake of contradiction, we assume that $f'(x_0) \neq 0$. Without loss of generality, we consider $f'(x_0) > 0$ as the case $f'(x_0) < 0$ is similar by symmetry. We have

$$f'(x_0) = \lim_{x \rightarrow x_0} \frac{f(x) - f(x_0)}{x - x_0} > 0.$$

Case 1: If f achieves maximum at x_0 , we take $(x_n) \rightarrow x_0$ with $x_n > x_0, \forall n \in \mathbb{N}$. Then the sequence

$$\left(\frac{f(x_n) - f(x_0)}{x_n - x_0} \right)_{n \in \mathbb{N}}$$

is convergent and converges to $f'(x_0)$. There exists some point $N \in \mathbb{N}$ such that

$$\frac{f(x_n) - f(x_0)}{x_n - x_0} > 0, \quad \forall n \geq N.$$

Since $x_n > x_0$, we infer that $f(x_n) > f(x_0), \forall n \geq N$. This contradicts with the maximality of $f(x_0)$.

Case 2: If f achieves minimum at x_0 , we take a sequence $(x_n) \rightarrow x_0$ with $x_n < x_0, \forall n \in \mathbb{N}$. Similarly, there exist some $M \in \mathbb{N}$ such that

$$\frac{f(x_n) - f(x_0)}{x_n - x_0} > 0, \quad \forall n \geq M.$$

It follows that $f(x_n) < f(x_0), \forall n \geq N$. This contradicts with minimality of $f(x_0)$.

In either case, we have a contradiction. Hence, $f'(x_0) = 0$ as desired. $\square$

Theorem. (Rolle's Theorem) Let $a < b$ be real numbers such that f is a continuous function on $[a, b]$ and differentiable on (a, b) . Assume further that $f(a) = f(b)$. Then, there exists at least one $x \in (a, b)$ such that $f'(x) = 0$.

Proof. By the Extreme Value Theorem, there exists $x_0, y_0 \in [a, b]$ such that

$$f(x_0) \leq f(x) \leq f(y_0), \quad \forall x \in [a, b].$$

If both x_0 and y_0 belong to $\{a, b\}$, then f is a constant function taking on the value $f(a) = f(b)$ throughout $[a, b]$. In this case, $f'(x) = 0$ for every $x \in (a, b)$, in particular, $f'(x_m) = 0$ where

$$x_m = \frac{a+b}{2} \in (a, b).$$

Otherwise, at least one of x_0 and y_0 belongs to (a, b) . So f attains at least one of its extrema at some point in $z_0 \in (a, b)$. From the previous theorem, we have $f'(z_0) = 0$. $\square$

Theorem. (Mean Value Theorem) Let $a < b$ be real numbers such that f is a continuous function on $[a, b]$ and differentiable on (a, b) . Then, there exists $x \in (a, b)$ such that

$$f'(x) = \frac{f(b) - f(a)}{b - a}.$$

Proof. We define $L(x)$ to be the straight line connecting $(a, f(a))$ and $(b, f(b))$. Its specific form is

$$L(x) = \frac{f(b) - f(a)}{b - a}(x - a) + f(a).$$

We now consider the function $g(x) = f(x) - L(x)$. Notice that g satisfies all the hypotheses of Rolle's theorem, so there exists some $x \in (a, b)$ such that

$$0 = g'(x) = f'(x) - L'(x) = f'(x) - \frac{f(b) - f(a)}{b - a}.$$

Hence, we get

$$f'(x) = \frac{f(b) - f(a)}{b - a}$$

for some $x \in (a, b)$. $\square$

Corollary. If f is differentiable on (a, b) , then

- (1) If $f'(x) > 0$, $\forall x \in (a, b)$, then f is strictly increasing.
- (2) If $f'(x) < 0$, $\forall x \in (a, b)$, then f is strictly decreasing.
- (3) If $f'(x) \geq 0$, $\forall x \in (a, b)$, then f is increasing.
- (4) If $f'(x) \leq 0$, $\forall x \in (a, b)$, then f is decreasing.

Proof. (1) For any $a < x_1 < x_2 < b$, we aim to prove that $f(x_1) < f(x_2)$. We apply the mean value theorem to f on $[x_1, x_2]$, there exists some $x_0 \in (x_1, x_2)$ such that

$$f'(x_0) = \frac{f(x_1) - f(x_2)}{x_1 - x_2} > 0$$

from where we get $f(x_1) < f(x_2)$.

The proofs of (2) (3) (4) are similar to (1) and thus we omit it here. $\square$

23 Lecture 23

The Riemann integral

In this section, we will formalize the idea of Riemann integration. The motivation is to compute the true area under the curves using modern analysis.

Notation. Let f be a bounded function on a closed interval $[a, b]$. For $S \subseteq [a, b]$, we use the notation

$$M(f, S) = \sup_{x \in S} f(x) = \sup\{f(x) : x \in S\} \quad \text{and} \quad m(f, S) = \inf_{x \in S} f(x) = \inf\{f(x) : x \in S\}.$$

A partition P of $[a, b]$ is a finite set P with the form

$$P = \{a = t_0 < t_1 < \cdots < t_n = b\}.$$

In other words, $P = \{t_0, t_1, \dots, t_n\}$ for some $n \in \mathbb{N}$ such that $a = t_0 < t_1 < \cdots < t_n = b$.

Definition. Let f be a bounded function on $[a, b]$. We define the upper Darboux sum $U(f, P)$ of f with respect to P as

$$U(f, P) = \sum_{k=1}^n M(f, [t_{k-1}, t_k]) \cdot (t_k - t_{k-1}).$$

Similarly, we define the lower Darboux sum $L(f, P)$ of f with respect to P as

$$L(f, P) = \sum_{k=1}^n m(f, [t_{k-1}, t_k]) \cdot (t_k - t_{k-1}).$$

Note. It is clear that the upper Darboux sum $U(f, P)$ overestimates the area under the curve while the lower Darboux sum $L(f, P)$ underestimates the area under the curve.

Proposition. Let f be bounded on $[a, b]$, then

$$L(f, P) \leq U(f, P)$$

for any partition P of $[a, b]$.

Proof. Let $P = \{a = t_0 < t_1 < \cdots < t_n = b\}$ be a partition of $[a, b]$. Thus, we have

$$L(f, P) = \sum_{k=1}^n m(f, [t_{k-1}, t_k]) \cdot (t_k - t_{k-1}) \leq \sum_{k=1}^n M(f, [t_{k-1}, t_k]) \cdot (t_k - t_{k-1}) = U(f, P),$$

as desired. □

The following lemma shows that we can lower the overestimate and raise the underestimate by making the partition finer.

Proposition. Let f be bounded on $[a, b]$. Suppose that P and Q are two partitions of $[a, b]$. Assume that $P \subseteq Q$, then

$$L(f, P) \leq L(f, Q) \leq U(f, Q) \leq U(f, P).$$

Note that $P \subseteq Q$ means that Q is a finer partition than P , i.e., all the partitioning points in P belong to Q .

Proof. The second inequality follows from the previous proposition. We aim to prove the first inequality and the third inequality is similar. Let $P = \{a = t_0 < t_1 < \cdots < t_n = b\}$. Use an induction argument (homework) we may assume that Q has only one more point, i.e., $Q = \{a = t_0 < t_1 < \cdots < t_{j-1} < u < t_j < \cdots < t_n = b\}$. Note that

$$L(f, P) = \sum_{k=1}^n m(f, [t_{k-1}, t_k]) \cdot (t_k - t_{k-1})$$

and

$$\begin{aligned} L(f, Q) &= \sum_{k=1}^{j-1} m(f, [t_{k-1}, t_k]) \cdot (t_k - t_{k-1}) + m(f, [t_{j-1}, u]) \cdot (u - t_{j-1}) + m(f, [u, t_j]) \cdot (t_j - u) \\ &\quad + \sum_{k=1}^{j+1} m(f, [t_{k-1}, t_k]) \cdot (t_k - t_{k-1}) \\ &\geq \sum_{k=1}^{j-1} m(f, [t_{k-1}, t_k]) \cdot (t_k - t_{k-1}) + m(f, [t_{j-1}, t_j]) \cdot (u - t_{j-1} + t_j - u) \\ &\quad + \sum_{k=1}^{j+1} m(f, [t_{k-1}, t_k]) \cdot (t_k - t_{k-1}) = L(f, P). \end{aligned}$$

Hence, the first inequality is proved. $\square$

Lemma. If f is a bounded function on $[a, b]$. Then for any partition P, Q of $[a, b]$, we have

$$L(f, P) \leq U(f, Q).$$

Proof. We consider the partition $P \cup Q$, which is finer than both P and Q . Using the above proposition, we have

$$L(f, P) \leq L(f, P \cup Q) \leq U(f, P \cup Q) \leq U(f, Q).$$

The lemma is thus proved. $\square$

Definition. Let f be a bounded function on $[a, b]$. The upper Darboux integral $U(f)$ of f over $[a, b]$ is defined as

$$U(f) = \inf\{U(f, P) : P \text{ is a partition of } [a, b]\}.$$

The lower Darboux integral $L(f)$ of f over $[a, b]$ is given by:

$$L(f) = \sup\{L(f, P) : P \text{ is a partition of } [a, b]\}.$$

We say that f is (Darboux) integrable on $[a, b]$ if

$$L(f) = U(f).$$

In this case, we define the (Darboux) integral of f as

$$\int_a^b f(x) dx = L(f) = U(f).$$

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Remark. If f is bounded on $[a, b]$, then $L(f) \leq U(f)$. This follows from the lemma from last lecture. Hence, to prove that f is integrable, we only need to prove that $L(f) \geq U(f)$. This motivates the following theorem.

Theorem. A bounded function f on $[a, b]$ is (Darboux) integrable if and only if

$$\text{for every } \epsilon > 0, \text{ there exists a partition } P \text{ of } [a, b] \text{ such that } U(f, P) - L(f, P) < \epsilon. \quad (*)$$

Proof. $\implies$: Suppose that f is integrable. Let $\epsilon > 0$. Since $L(f)$ is the supremum and $U(f)$ is the infimum, there exists partitions P_1 and P_2 such that

$$L(f, P_1) > L(f) - \frac{\epsilon}{2} \quad \text{and} \quad U(f, P_2) < U(f) + \frac{\epsilon}{2}.$$

For $P = P_1 \cup P_2$, we apply the lemma from last lecture to get

$$U(f, P) - L(f, P) \leq U(f, P_2) - L(f, P_1) < U(f) + \frac{\epsilon}{2} - (L(f) - \frac{\epsilon}{2}) = U(f) - L(f) + \epsilon.$$

Since f is integrable, by definition we have $U(f) = L(f)$. Thus, $(*)$ holds.

$\impliedby$: Suppose that $(*)$ holds. Let $\epsilon > 0$, there exists a partition P such that $U(f, P) - L(f, P) < \epsilon$. Then

$$U(f) \leq U(f, P) = U(f, P) - L(f, P) + L(f, P) < \epsilon + L(f, P) \leq \epsilon + L(f).$$

Since $\epsilon > 0$ is arbitrary, we conclude that $U(f) \leq L(f)$ and hence $U(f) = L(f)$. $\square$

Example. Let $f(x) = C$ on $[a, b]$ for some constant C . Prove that f is integrable on $[a, b]$.

Proof. Let $\epsilon > 0$. We take the partition $P = \{a = t_0 < t_1 = b\}$. Then

$$U(f, P) = C \cdot (b - a) \quad \text{and} \quad L(f, P) = C \cdot (b - a).$$

Hence, $U(f, P) - L(f, P) = 0 < \epsilon$. By the theorem, f is integrable on $[a, b]$. $\square$

Example. Let $f(x) = x$ on $[0, 1]$. Prove that f is integrable on $[0, 1]$.

Proof. Let $\epsilon > 0$. We take some $n \in \mathbb{N}$ such that $n > \frac{1}{\epsilon}$. For the partition $P = \{0 < \frac{1}{n} < \frac{2}{n} < \dots < \frac{n-1}{n} < 1\}$, we have

$$U(f, P) = \frac{1}{n} \cdot \frac{1}{n} + \frac{1}{n} \cdot \frac{2}{n} + \dots + \frac{1}{n} \cdot \frac{n}{n} = \frac{1}{n^2}(1 + 2 + \dots + n) = \frac{n(n+1)}{2n^2}$$

and

$$L(f, P) = \frac{1}{n} \cdot \frac{1}{n} + \frac{1}{n} \cdot \frac{2}{n} + \dots + \frac{1}{n} \cdot \frac{n-1}{n} = \frac{1}{n^2}(1 + 2 + \dots + (n-1)) = \frac{n(n-1)}{2n^2}.$$

Therefore,

$$U(f, P) - L(f, P) = \frac{2n}{2n^2} = \frac{1}{n} < \epsilon.$$

By the theorem, f is integrable on $[0, 1]$. $\square$

Example. Consider the function on $[0, 1]$ defined by

$$f(x) = \begin{cases} 1, & x \in \mathbb{Q} \\ 0, & x \notin \mathbb{Q}. \end{cases}$$

Prove that f is not integrable on $[0, 1]$.

Proof. For the sake of contradiction, we assume that f is integrable. Let $\epsilon = \frac{1}{2}$, there exists a partition $P = \{0 = t_0 < t_1 < \dots < t_n = 1\}$ such that $U(f, P) - L(f, P) < \epsilon$. Since rationals and irrationals are dense, we get

$$M(f, [t_{k-1}, t_k]) = \sup\{0, 1\} = 1 \quad \text{and} \quad m(f, [t_{k-1}, t_k]) = \inf\{0, 1\} = 0.$$

Thus,

$$U(f, P) = \sum_{k=1}^n 1 \cdot (t_k - t_{k-1}) = 1, \quad L(f, P) = \sum_{k=1}^n 0 \cdot (t_k - t_{k-1}) = 0.$$

This contradicts with $U(f, P) - L(f, P) < \frac{1}{2}$. □

The mesh of a partition P is the maximum length of the subintervals in P . For example, if $P = \{a = t_0 < t_1 < \dots < t_n = b\}$, then $\text{mesh}(P) = \max\{t_k - t_{k-1} : k = 1, 2, \dots, n\}$.

Theorem. A bounded function on $[a, b]$ is (Darboux) integrable if and only if

$$\forall \epsilon > 0, \exists \delta > 0 \text{ such that } \text{mesh}(P) < \delta \implies U(f, P) - L(f, P) < \epsilon.$$

Proof. $\Leftarrow$: This follows from the previous theorem.

$\Rightarrow$: Suppose that f is integrable on $[a, b]$. Let $\epsilon > 0$. From the previous theorem, there exists a partition $P_0 = \{a = u_0 < u_1 < \dots < u_m = b\}$ such that

$$U(f, P_0) - L(f, P_0) < \frac{\epsilon}{2}.$$

Since f is bounded, there exists some $M > 0$ such that $|f(x)| \leq M$ for every $x \in [a, b]$. Let $\delta = \epsilon/8mM > 0$. Fix a partition $P = \{a = t_0 < t_1 < \dots < t_n = b\}$ such that $\text{mesh}(P) < \delta$. Define $Q = P \cup P_0$. If it is the case that Q has one more element than P , then by re-examining the proof of the proposition from last lecture, we see that

$$L(f, Q) - L(f, P) \leq M \cdot \text{mesh}(P) - (-M) \cdot \text{mesh}(P) = 2M \cdot \text{mesh}(P).$$

Since Q has at most m elements that are not in P , an induction argument yields

$$L(f, Q) - L(f, P) \leq m \cdot (2M \cdot \text{mesh}(P)) < \frac{\epsilon}{4}.$$

Since Q is finer than P_0 , we get

$$L(f, P_0) - L(f, P) \leq L(f, Q) - L(f, P) < \frac{\epsilon}{4}.$$

Similarly, by examining the upper Darboux sum, we see

$$U(f, P) - U(f, P_0) < \frac{\epsilon}{4}.$$

Combining these two inequalities, we obtain

$$U(f, P) - L(f, P) < U(f, P_0) - L(f, P_0) + \frac{\epsilon}{2} < \epsilon,$$

as desired. □

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Definition. Let f be a bounded function on $[a, b]$ and P be a partition of $[a, b]$. A Riemann sum of f is defined as

$$S = \sum_{k=1}^n f(x_k) \cdot (t_k - t_{k-1})$$

where $P = \{a = t_0 < t_1 < \dots < t_n = b\}$ and $x_k \in [t_{k-1}, t_k]$ for $k = 1, \dots, n$.

Definition. We say that f is Riemann integrable on $[a, b]$ if there exists a real number r with the following property

$$\forall \epsilon > 0, \exists \delta > 0, \text{ such that } \text{mesh}(P) < \delta \implies |S - r| < \epsilon.$$

The number r is the Riemann integral of f on $[a, b]$ and is written as $r = \int_a^b f(x)dx$.

Informal discussion. The Darboux integral and the Riemann integral are two different ways of looking at the same idea – measuring the area under a curve. Even though they have different definitions, they turn out to be completely equivalent.

Theorem. A bounded function f on $[a, b]$ is Riemann integrable if and only if it is Darboux integrable, in which case the values of the integrals agree.

Corollary. Let f be a bounded and integrable function on $[a, b]$. Suppose (S_n) is a sequence of Riemann sums, with corresponding partitions P_n , satisfying $\lim_{n \rightarrow \infty} \text{mesh}(P_n) = 0$. Then

$$\int_a^b f = \lim_{n \rightarrow \infty} S_n.$$

Proof. Let $\epsilon > 0$. By definition, there exists some $\delta > 0$ such that

$$\text{mesh}(P) < \delta \implies \left| S - \int_a^b f \right| < \epsilon.$$

Since $\lim_{n \rightarrow \infty} \text{mesh}(P_n) = 0$, there exists some $N \in \mathbb{N}$ such that $|\text{mesh}(P_n) - 0| < \delta$, for $n \geq N$. For the Riemann sums S_n associated with P_n , we have

$$\left| S_n - \int_a^b f \right| < \epsilon, \quad \forall n \geq N.$$

Since ϵ is arbitrary, we get $\lim_{n \rightarrow \infty} S_n = \int_a^b f$. □

Example. Find the integral of $f(x) = x$ on $[0, 1]$.

Proof. From last time, we have shown that f is integrable. Let $n \in \mathbb{N}$, we take $P_n = \{0 < \frac{1}{n} < \dots < \frac{n-1}{n} < 1\}$ and the sample points $x_k = t_k$ for all $k = 1, \dots, n$. Then

$$S_n(f, P) = U_n(f, P) = \frac{n+1}{2n}.$$

Since $\text{mesh}(P_n) = \frac{1}{n}$ and it is clear that $\lim_{n \rightarrow \infty} \text{mesh}(P_n) \rightarrow 0$, from the corollary we obtain

$$\int_0^1 f = \lim_{n \rightarrow \infty} S_n = \frac{1}{2}.$$

□

Properties of the Riemann integral

Theorem. Every monotone function f on $[a, b]$ is integrable.

Proof. Without loss of generality, we assume that f is increasing on $[a, b]$. If $f(a) = f(b)$, then f would be constant and thus integrable. Now, we consider $f(a) < f(b)$. For any $x \in [a, b]$, we have $f(a) \leq f(x) \leq f(b)$ so the function f is bounded on $[a, b]$. Let $\epsilon > 0$ and we take

$$\delta = \frac{\epsilon}{f(b) - f(a)}.$$

Then, for any partition $P = \{a = t_0 < t_1 < \dots < t_n = b\}$ with $\text{mesh}(P) < \delta$, we get

$$\begin{aligned} U(f, P) - L(f, P) &= \sum_{k=1}^n (M(f, [t_{k-1}, t_k]) - m(f, [t_{k-1}, t_k])) (t_k - t_{k-1}) \\ &= \sum_{k=1}^n (f(t_k) - f(t_{k-1}))(t_k - t_{k-1}) \\ &< \delta \sum_{k=1}^n (f(t_k) - f(t_{k-1})) = \delta(f(b) - f(a)) = \epsilon. \end{aligned}$$

Thus, $U(f, P) - L(f, P) < \epsilon$, proving that f is integrable.

□

Theorem. Every continuous function f on $[a, b]$ is integrable.

Proof. Let $\epsilon > 0$. Since f is uniformly continuous on $[a, b]$, there exists some $\delta > 0$ such that

$$|x - y| < \delta \implies |f(x) - f(y)| < \frac{\epsilon}{b - a}.$$

Fix a partition $P = \{a = t_0 < t_1 < \dots < t_n = b\}$ with $\text{mesh}(P) < \delta$. By the extreme value theorem, we have

$$M(f, [t_{k-1}, t_k]) - m(f, [t_{k-1}, t_k]) < \frac{\epsilon}{b - a},$$

for each $k = 1, \dots, n$. Hence,

$$U(f, P) - L(f, P) = \sum_{k=1}^n (M(f, [t_{k-1}, t_k]) - m(f, [t_{k-1}, t_k]))(t_k - t_{k-1}) < \frac{\epsilon}{b - a} \sum_{k=1}^n (t_k - t_{k-1}) = \epsilon.$$

By the theorem, f is integrable on $[a, b]$.

□

Theorem. If f and g are integrable on $[a, b]$, then

(1) cf is integrable and $\int_a^b(cf) = c \int_a^b f$, for any constant $c \in \mathbb{R}$.

(2) $f \pm g$ is integrable and

$$\int_a^b (f \pm g) = \int_a^b f \pm \int_a^b g.$$

(3) fg is integrable.

Example. Prove that piecewise continuous function is integrable.

Proof. Let f be piecewise continuous on $[a, b]$. We first assume that f has two pieces, i.e.,

$$f(x) = \begin{cases} g(x), & x \in [a, c) \\ h(x), & x \in [c, b] \end{cases}$$

for some $c \in (a, b)$, where g and h are continuous on $[a, c)$ and $[c, b]$, respectively. Then we may use an induction argument to prove the general case.

Let $\epsilon > 0$. On the interval $[a, c]$, we know g is continuous on $[a, c)$ and $x = c$ is a single point, we get that f is integrable on $[a, c]$. There exists a partition P of $[a, c]$ such that

$$U(g, P) - L(g, P) < \frac{\epsilon}{2}.$$

Similarly, there exists a partition Q of $[c, b]$ such that

$$U(h, Q) - L(h, Q) < \frac{\epsilon}{2}.$$

Let $R = P \cup Q$. Then

$$U(f, R) = U(g, P) + U(h, Q) \quad \text{and} \quad L(f, R) = L(g, P) + L(h, Q).$$

Therefore,

$$U(f, R) - L(f, R) < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon.$$

Therefore, f is integrable on $[a, b]$ by the theorem. □